📈 CELL 11+: COX ANALYSIS & PLOTTING

Competing Risks

Cumulative Incidence Function Curve (empirical)

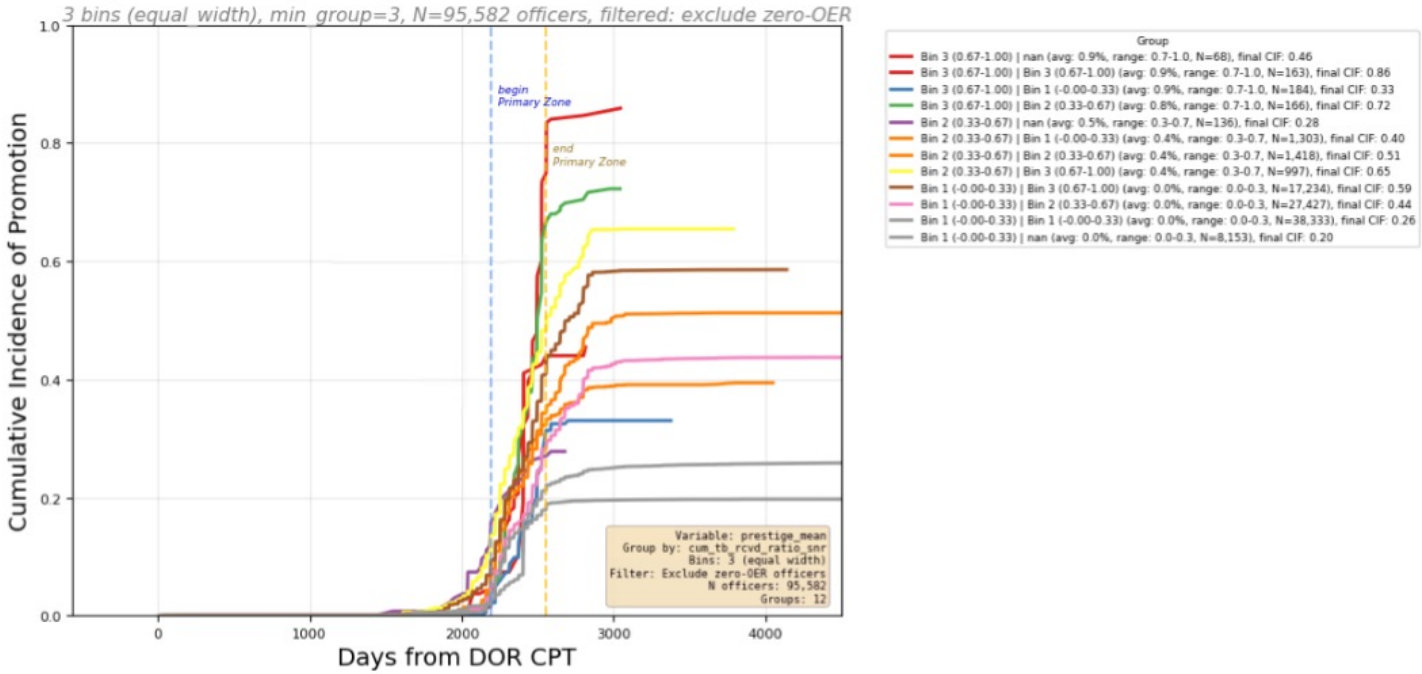
$prestige_mean$

Grouped by

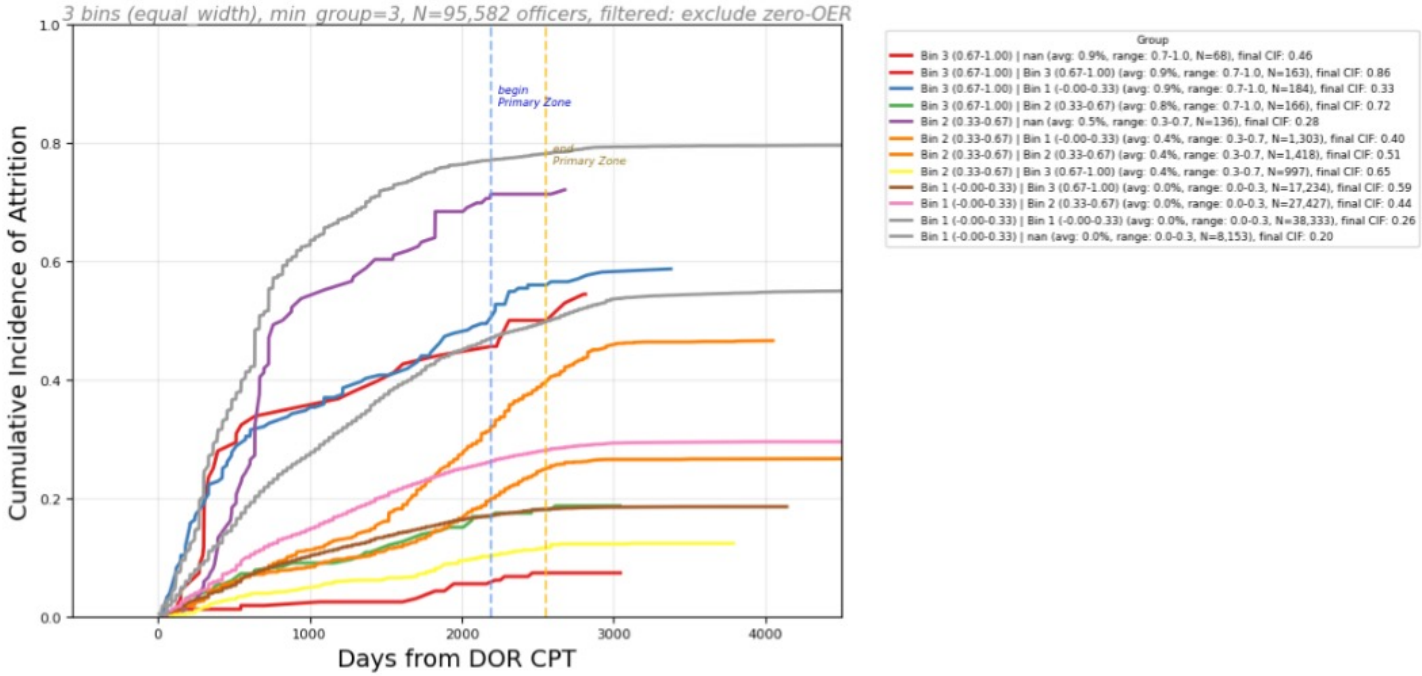
$cum_tb_rcvd_ratio_snr$

• Group 'nan | Bin 1 (-0.00-0.33)': Attrition CIF final=0.798, actual rate=0.798 (6,506/8,153)

Promotion Probability (CIF): Competing Risks: Prestige Mean by Cum Tb Rcvd Ratio Snr



Attrition Probability (CIF): Competing Risks: Prestige Mean by Cum Tb Rcvd Ratio Snr



✅ Saved: /csv/csv_plots/cp_prest_mean_cumtbrcvdr_b3_ev_noOER.png

📈 CELL 11+: COX ANALYSIS & PLOTTING

Competing Risks

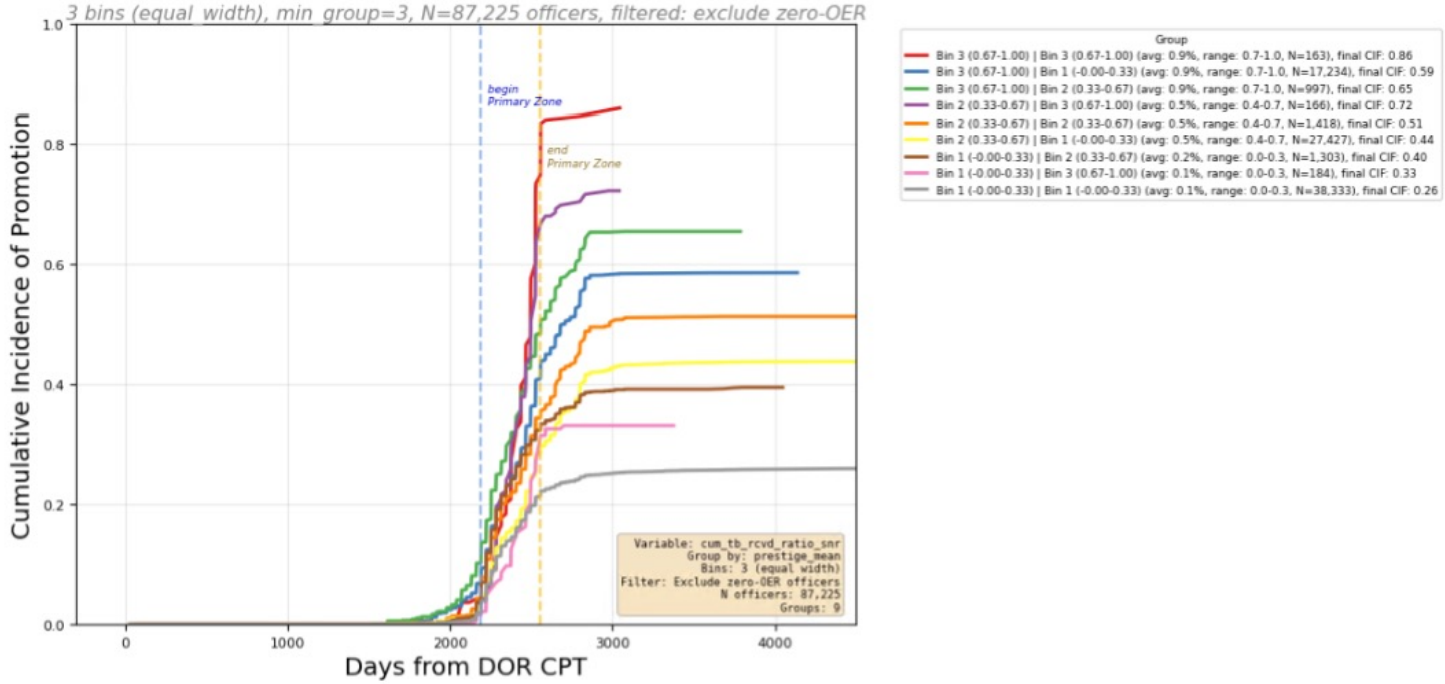
Cumulative Incidence Function Curve (empirical)

$cum_tb_rcvd_ratio_snr$

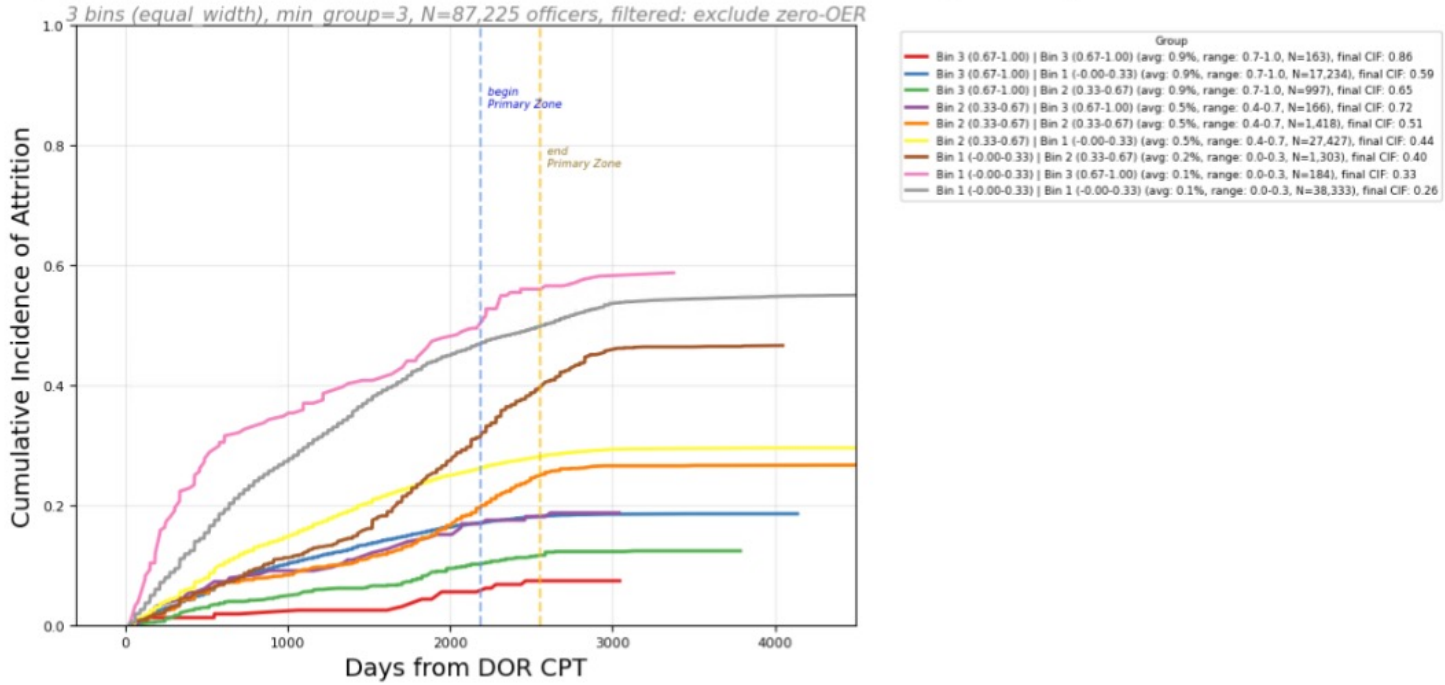
Grouped by

$prestige_mean$

Promotion Probability (CIF): Competing Risks: Cumulative TB Rcvd Ratio SNR by Prestige Mean



Attrition Probability (CIF): Competing Risks: Cumulative TB Rcvd Ratio SNR by Prestige Mean



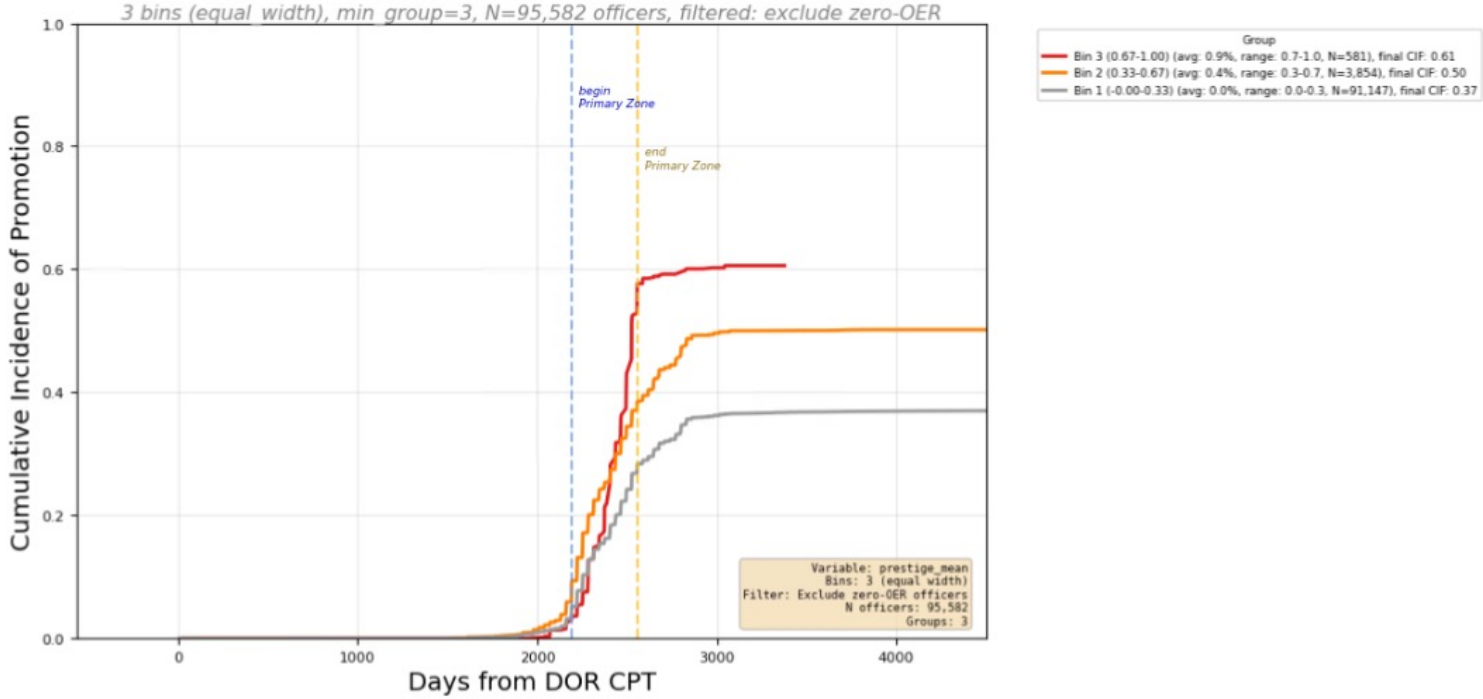
📈 CELL 11+: COX ANALYSIS & PLOTTING

Competing Risks

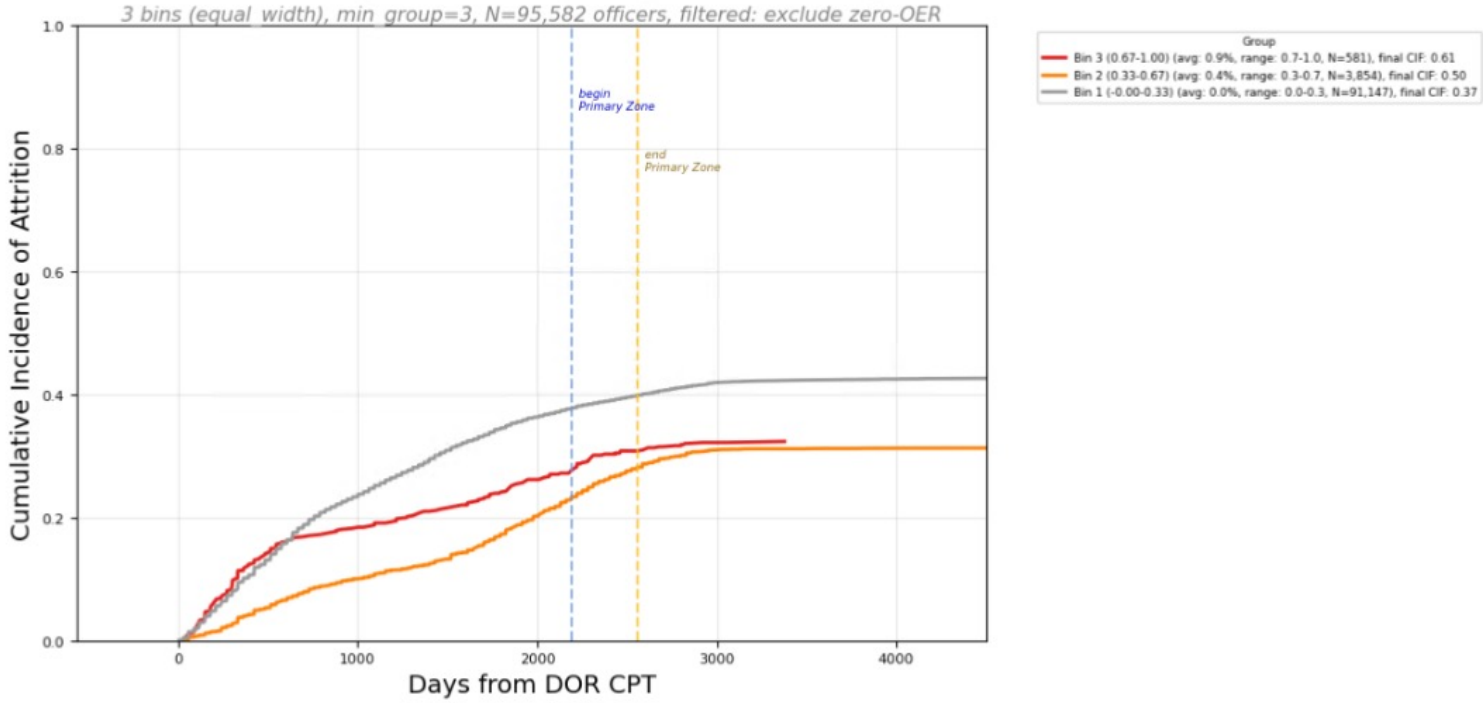
Cumulative Incidence Function Curve (empirical)

$prestige_mean$

Promotion Probability (CIF): Competing Risks: Prestige Mean



Attrition Probability (CIF): Competing Risks: Prestige Mean



=== 📦 12.6. PARTIAL EFFECTS PLOTS ===

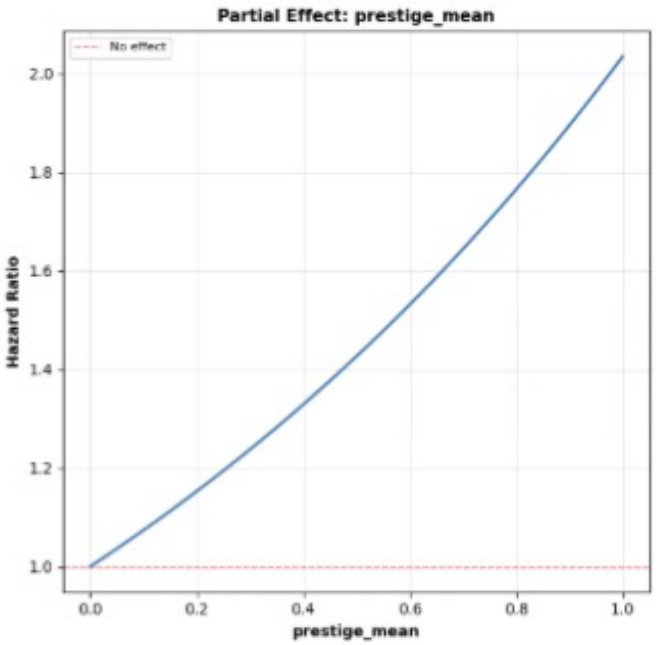
Partial Effects Plots for

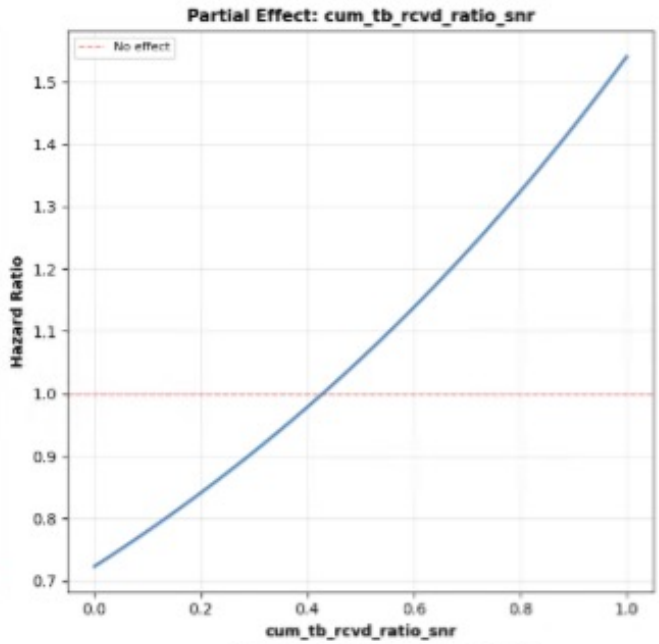
$prestige_mean$ and $cum_tb_rcvd_ratio_snr$

As well as

$z_prestige_mean$ and $z_cum_tb_rcvd_ratio_snr$

=== 📦 12.6. PARTIAL EFFECTS PLOTS ===



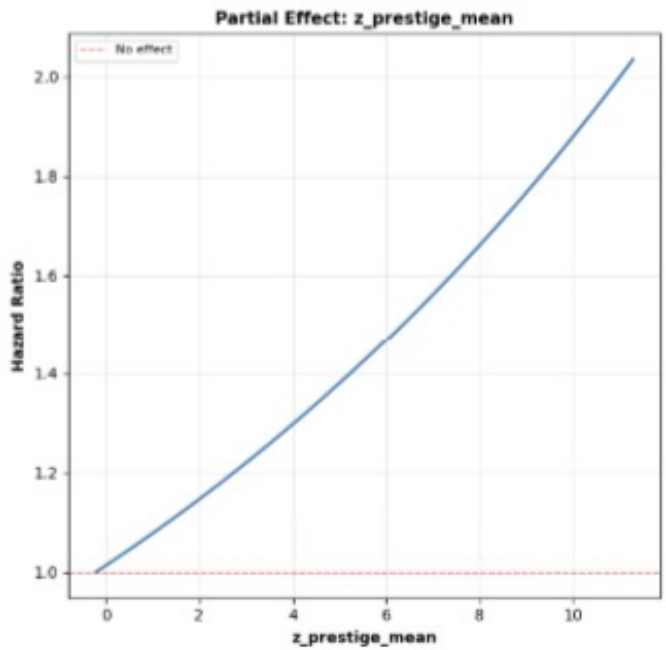


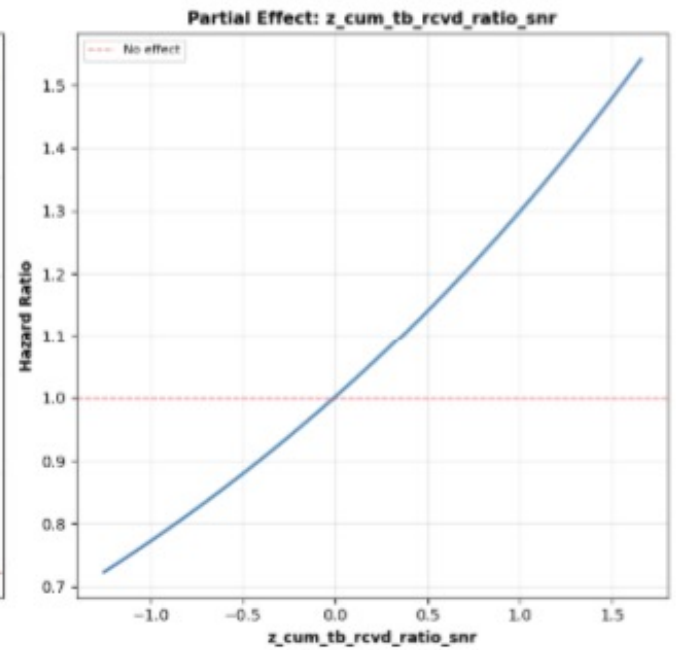
$prestige_mean$ and

$cum_tb_rcvd_ratio_snr$

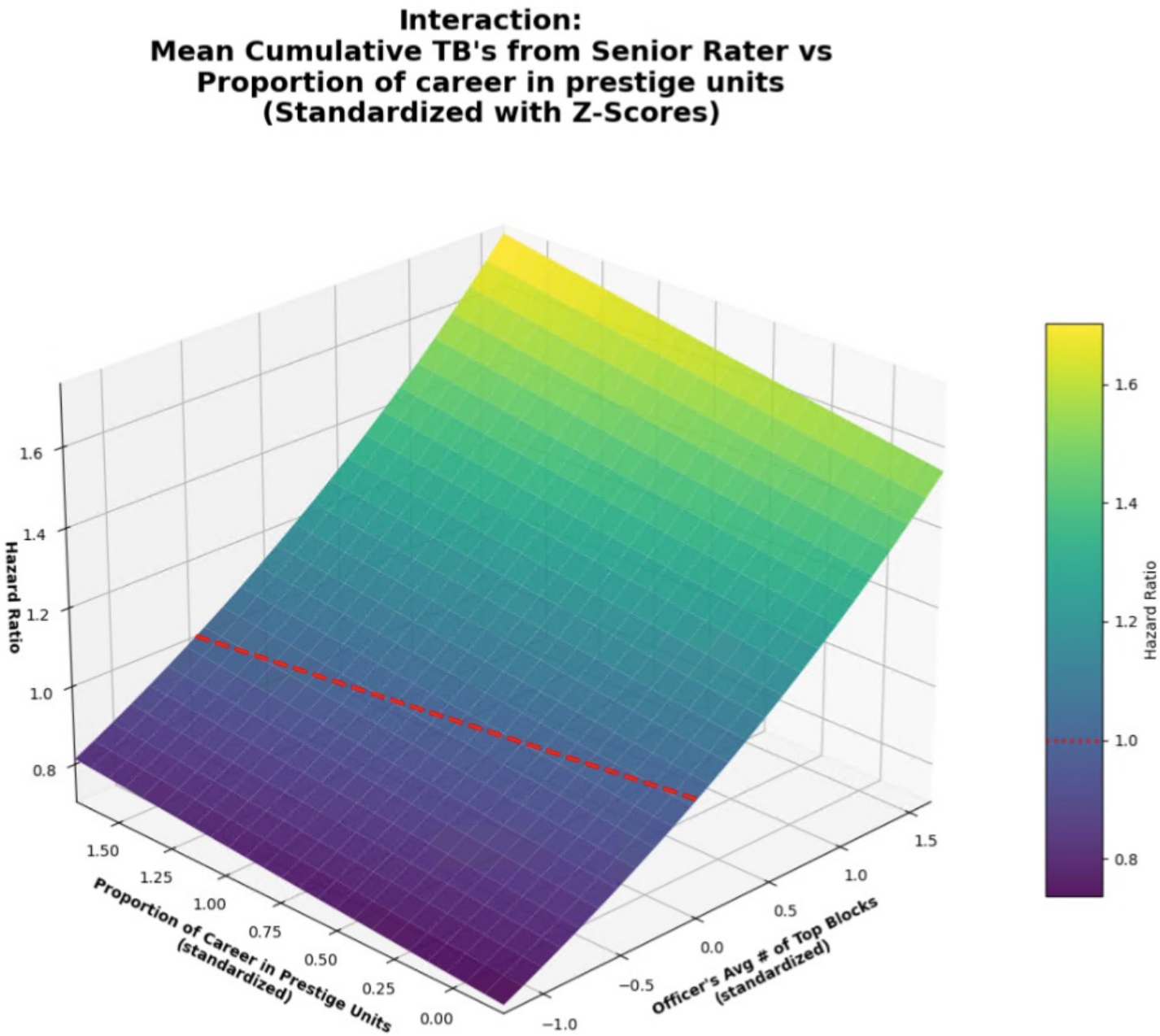
$z_prestige_mean$ and

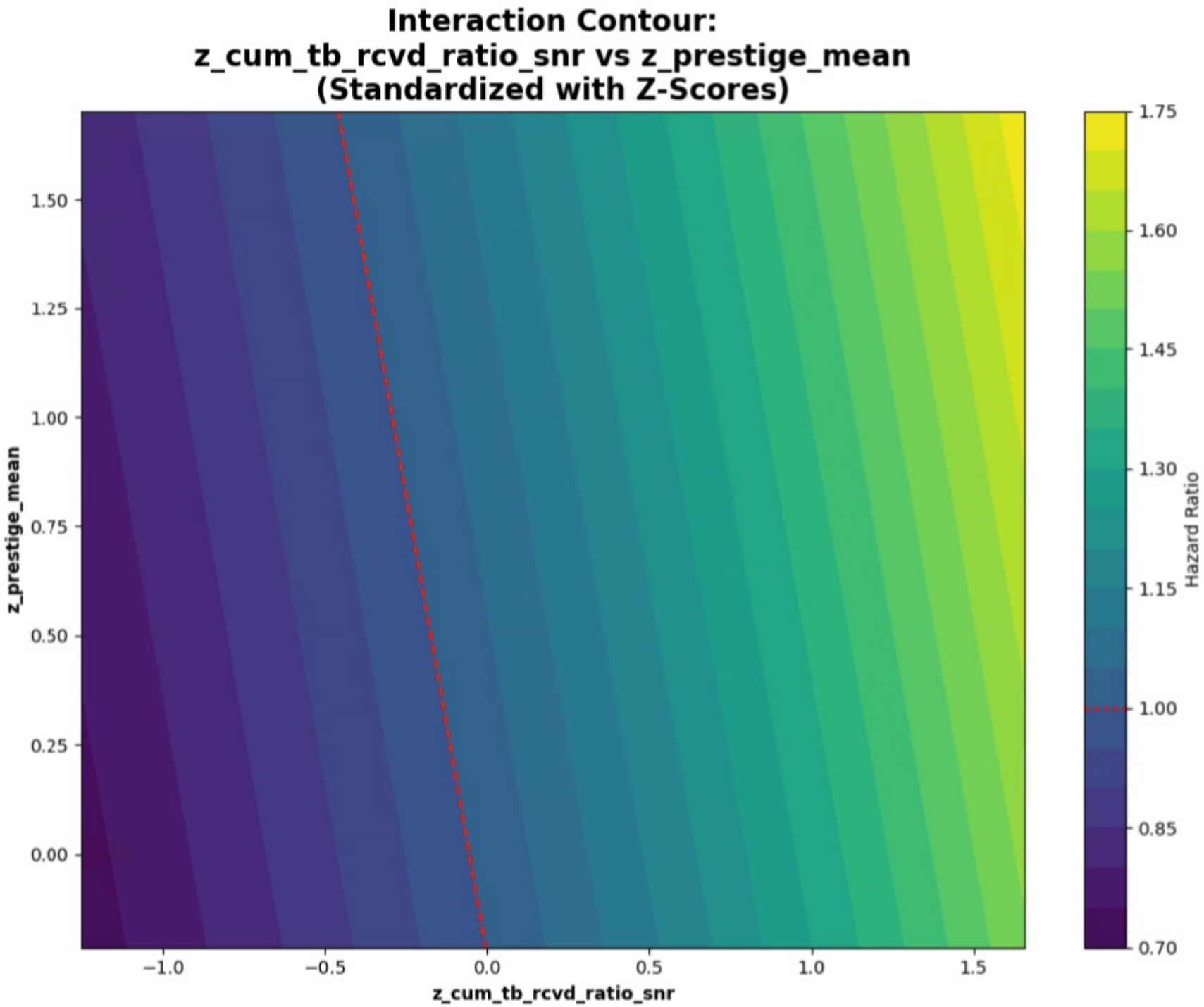
$z_cum_tb_rcvd_ratio_snr$





=== 📦 12.7. INTERACTION EFFECTS (3D PLOTS) ===




Interaction Contour:
z_cum_tb_rcvd_ratio_snr vs z_prestige_mean
(Standardized with Z-Scores)
z_prestige_mean
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−1.0
−0.5
0.0
0.5
1.0
1.5
z_cum_tb_rcvd_ratio_snr
Hazard Ratio
1.75
1.60
1.45
1.30
1.15
1.00
0.85
0.70

=== 📦 12.3. FIT FULL MODEL (STATIC + TIME-VARYING) ===

A	B	C
variable	coefficient	signal_ratio
sex	0.12334502006847842	1.1312746661078021
z_prestige_mean	0.06176310580886882	1.0637103280808038
z_cum_tb_rcvd_ratio_snr	0.26006484482171984	1.2970401291359588

To quantify the two effects with respect to each other:

If the **fitted model** is

$$\log(HR) = b1 * x + b2 * y + b3 * x * y$$

with $x = z_cum_tb_rcvd_ratio_snr$ and $y = z_prestige_mean$,

then:

Marginal effect of x at a given y: $\frac{\partial}{\partial x}\log(HR) = b1 + b3 * y$

Marginal effect of y at a given x: $\frac{\partial}{\partial y}\log(HR) = b2 + b3 * x$

A simple **trade-off rate** (how much x you need to change to offset y) is therefore:

$$\frac{dx}{dy} = -\frac{(b2 + b3 * x)}{(b1 + b3 * y)}$$

If there's **no interaction** $(b3 = 0)$, then it simplifies to: $\frac{dx}{dy} = -\frac{b2}{b1}$

So, a $+1\ \sigma$ increase in $z_prestige_mean$

is equivalent to

a $-\frac{b2}{b1}\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$.

Using the generated coefficient table:

A	B	C
variable	coefficient	signal_ratio
sex	0.12334502008847842	1.1312746661078021
z_prestige_mean	0.06176310580886882	1.0637103280808038
z_cum_tb_rcvd_ratio_snr	0.26008484482171984	1.2970401291359588

$b_{cum} = 0.2601$ for $z_cum_tb_rcvd_ratio_snr$
$b_{prestige} = 0.0618$ for $z_prestige_mean$

With no interaction term, the trade-off at constant HR is:

$$\frac{dx}{dy} = -\frac{b_{prestige}}{b_{cum}} = -0.2375$$

Interpretation (z-score units):

A $+ 1\ \sigma$ change in $z_prestige_mean$
is offset by
A $- 0.2375\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$
to keep HR constant.

Equivalently:

A $+ 1\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$

is about $4.21\times$

the effect of A $+ 1\ \sigma$ change in $z_prestige_mean$

Since

$$\frac{b_{cum}}{b_{prestige}} \approx 4.21$$

📈 CELL 11+: COX ANALYSIS & PLOTTING

The following results are counterintuitive:

We can see while plotting Competing Risks Cumulative Incidence Function Curves (empirical) that :

$tb_ratio_rank_norm_snr$

is positively correlated with promotion.

However, when fitted with $cum_tb_rcvd_ratio_snr$,

using a Cox Proportional Hazards model,

While looking at Partial Effects, we will see that correlation ***reverses***.

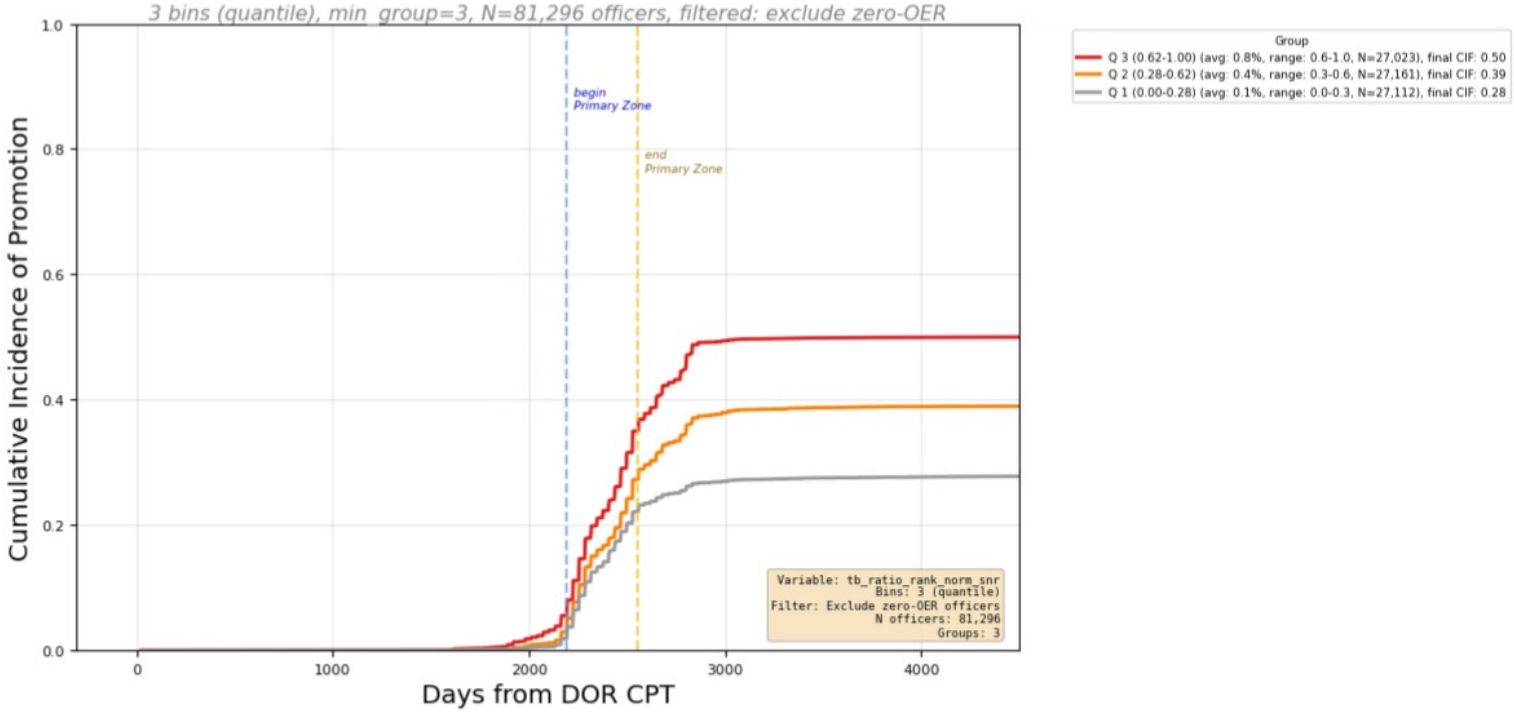
📈 CELL 11+: COX ANALYSIS & PLOTTING

Competing Risks Plots

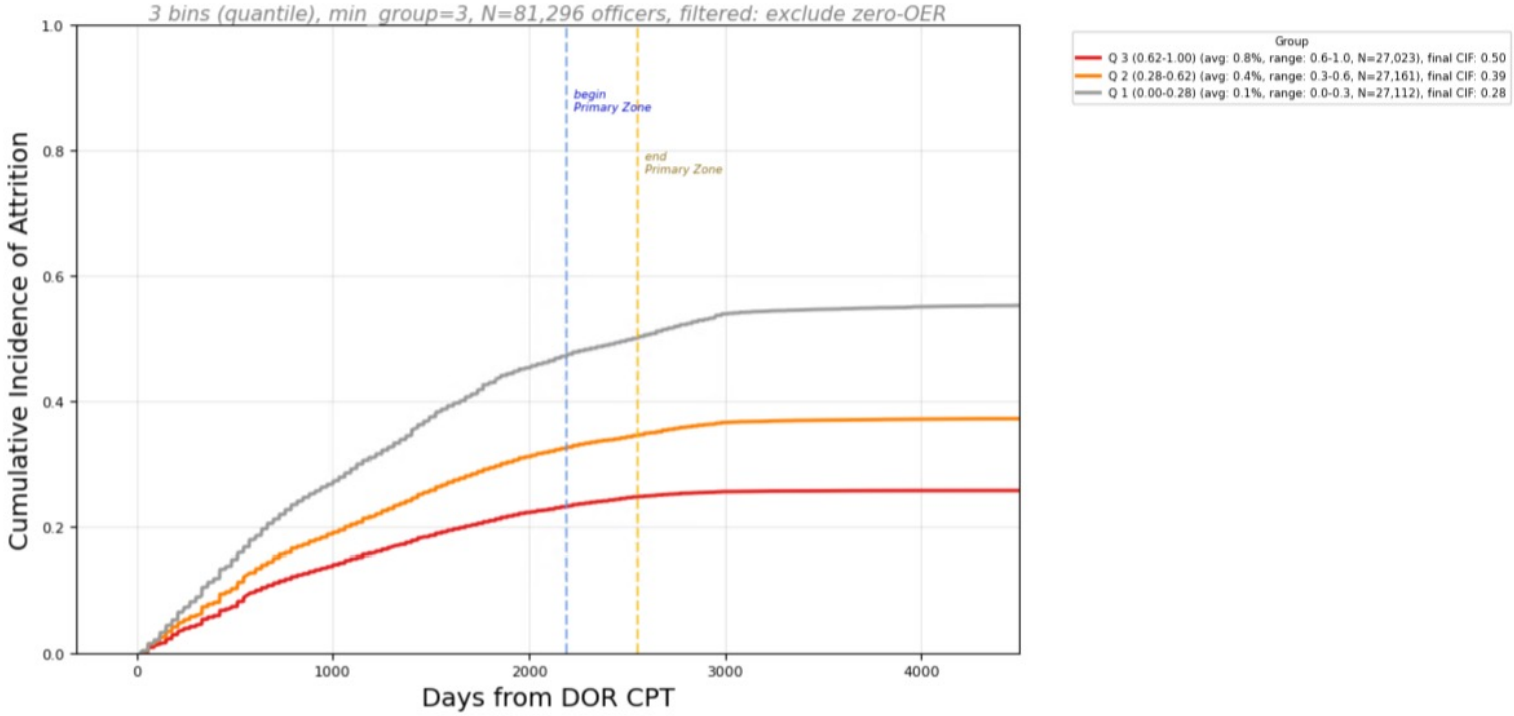
Cumulative Incidence Function Curve (empirical)

$tb_ratio_rank_norm_snr$

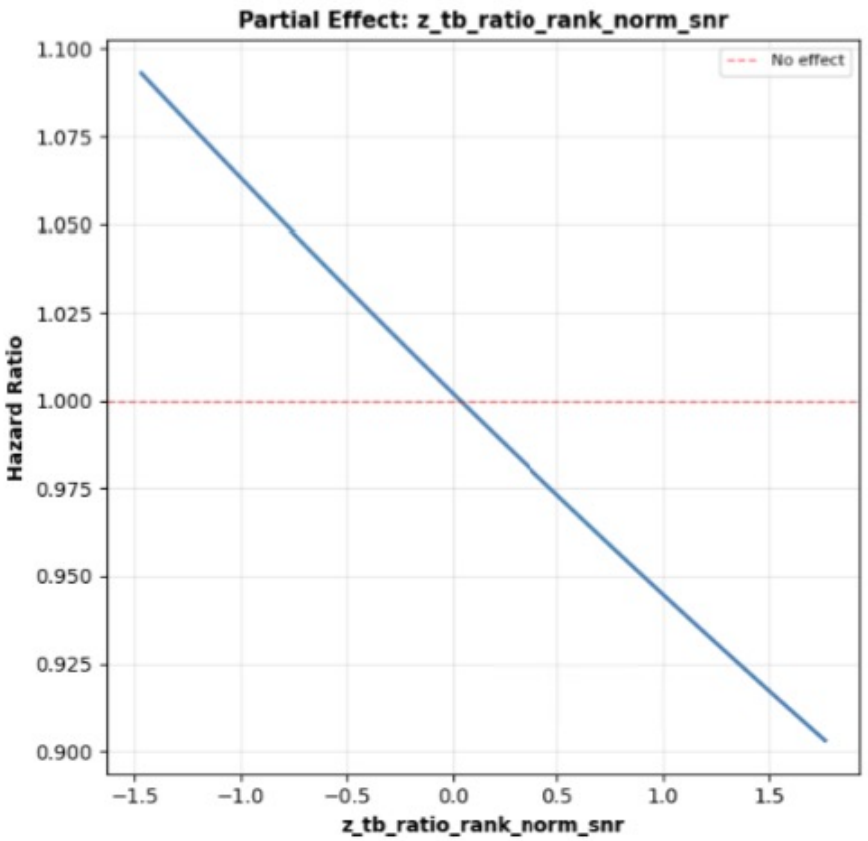
Promotion Probability (CIF): Competing Risks: TB Ratio Rank Norm SNR

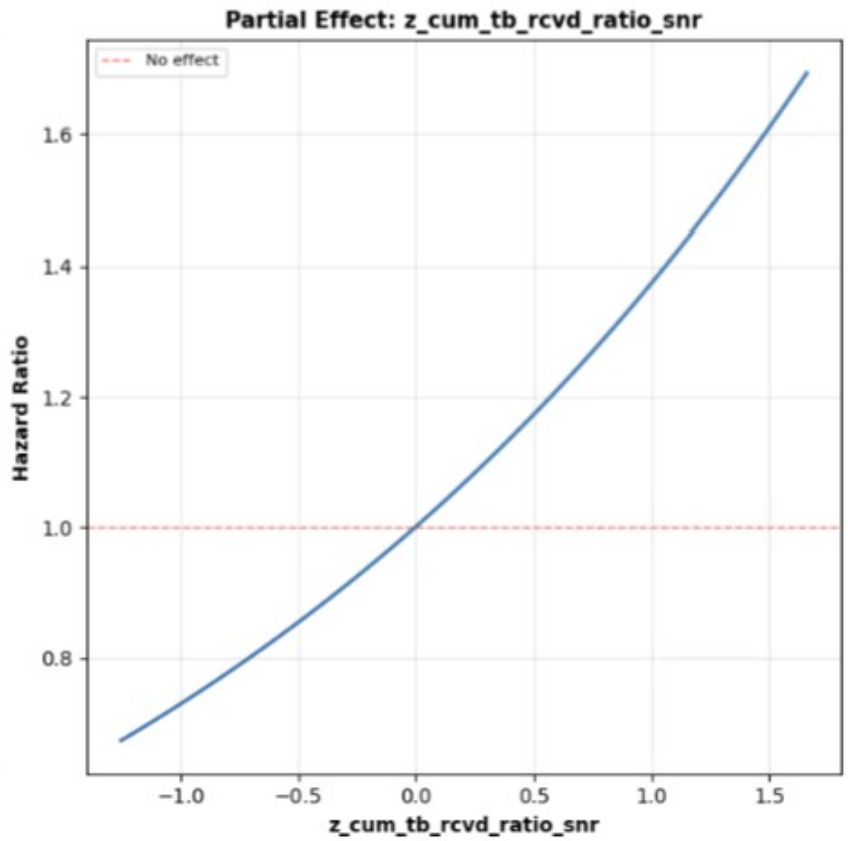


Attrition Probability (CIF): Competing Risks: TB Ratio Rank Norm SNR

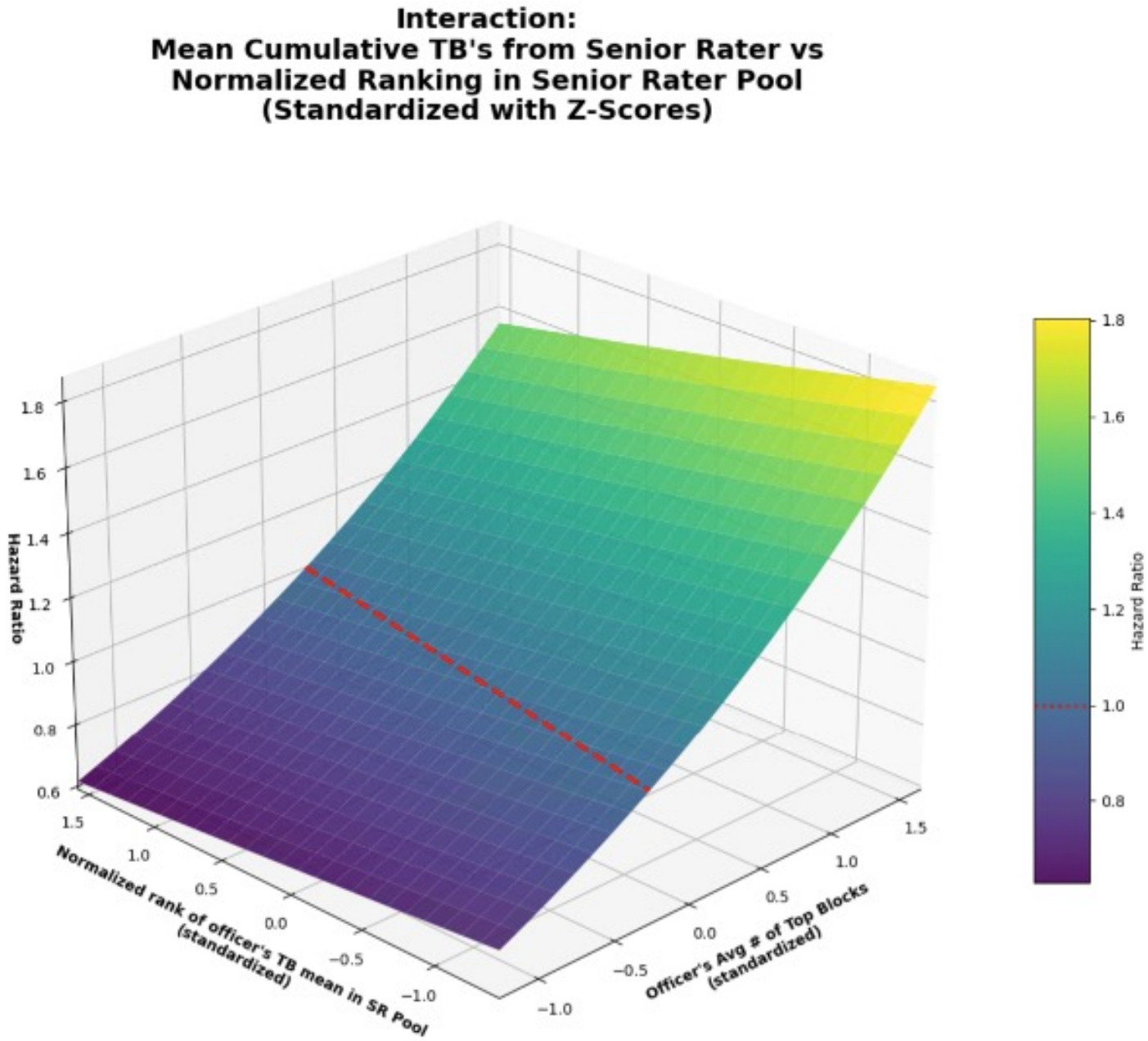


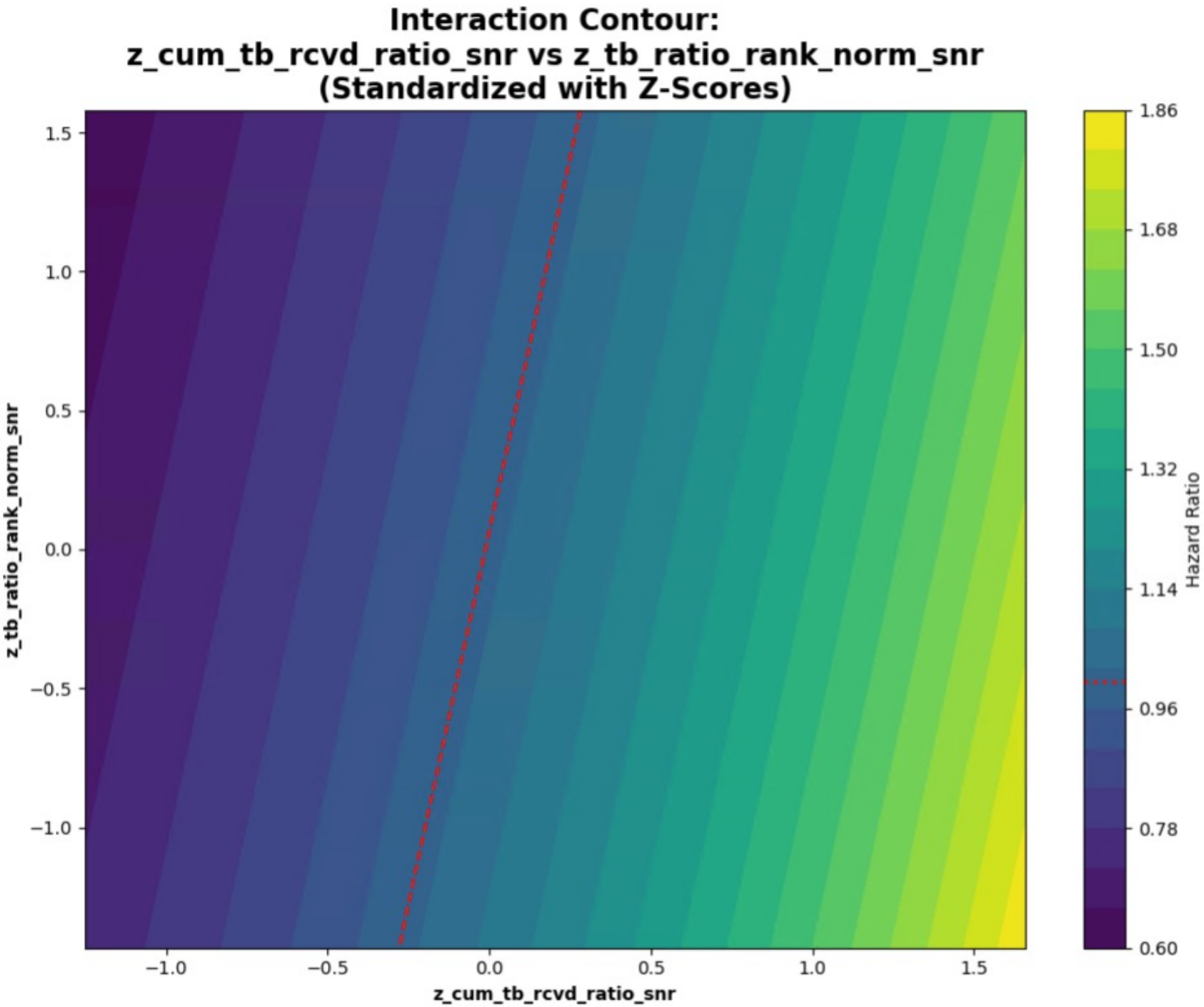
=== 📦 12.6. PARTIAL EFFECTS PLOTS ===





=== 📦 12.7. INTERACTION EFFECTS (3D PLOTS) ===




Interaction Contour:
z_cum_tb_rcvd_ratio_snr vs z_tb_ratio_rank_norm_snr
(Standardized with Z-Scores)
z_tb_ratio_rank_norm_snr
z_cum_tb_rcvd_ratio_snr
Hazard Ratio
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
1.86
1.68
1.50
1.32
1.14
0.96
0.78
0.60

=== 📦 12.3. FIT FULL MODEL (STATIC + TIME-VARYING) ===

A	B	C
variable	coefficient	signal_ratio
sex	0.15599809946367996	1.1688239816953292
z_tb_ratio_rank_norm_snr	-0.05903922283413774	0.9426697942509876
z_cum_tb_rcvd_ratio_snr	0.3169358418324863	1.3729144854222168

To quantify the two effects with respect to each other:

If the **fitted model** is

$$\log(HR) = b1 * x + b2 * y + b3 * x * y$$

with $x = z_cum_tb_rcvd_ratio_snr$ and $y = z_tb_ratio_rank_norm_snr$,

then:

Marginal effect of x at a given y: $\frac{\partial}{\partial x}\log(HR) = b1 + b3 * y$

Marginal effect of y at a given x: $\frac{\partial}{\partial y}\log(HR) = b2 + b3 * x$

A simple **trade-off rate** (how much x you need to change to offset y) is therefore:

$$\frac{dx}{dy} = -\frac{(b2 + b3 * x)}{(b1 + b3 * y)}$$

If there's **no interaction** $(b3 = 0)$, then it simplifies to: $\frac{dx}{dy} = -\frac{b2}{b1}$

So, a $+1\ \sigma$ increase in $z_tb_ratio_rank_norm_snr$

is equivalent to

a $-\frac{b2}{b1}\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$.

Using the generated coefficient table:

A	B	C
variable	coefficient	signal_ratio
sex	0.15599809946367996	1.1688239816953292
z_tb_ratio_rank_norm_snr	-0.05903922283413774	0.9426697942509876
z_cum_tb_rcvd_ratio_snr	0.3169358418324863	1.3729144854222168

$b_{cum} = 0.3169$ for $z_cum_tb_rcvd_ratio_snr$
$b_{rank} = -0.0590$ for $z_tb_ratio_rank_norm_snr$

With no interaction term, the trade-off at constant HR is:

$$\frac{dx}{dy} = -\frac{b_{rank}}{b_{cum}} = 0.1863$$

Interpretation (z-score units):

A $+1\ \sigma$ change in $z_tb_ratio_rank_norm_snr$
is offset by
A $+0.1863\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$
to keep HR constant.

Equivalently:

A $+1\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$

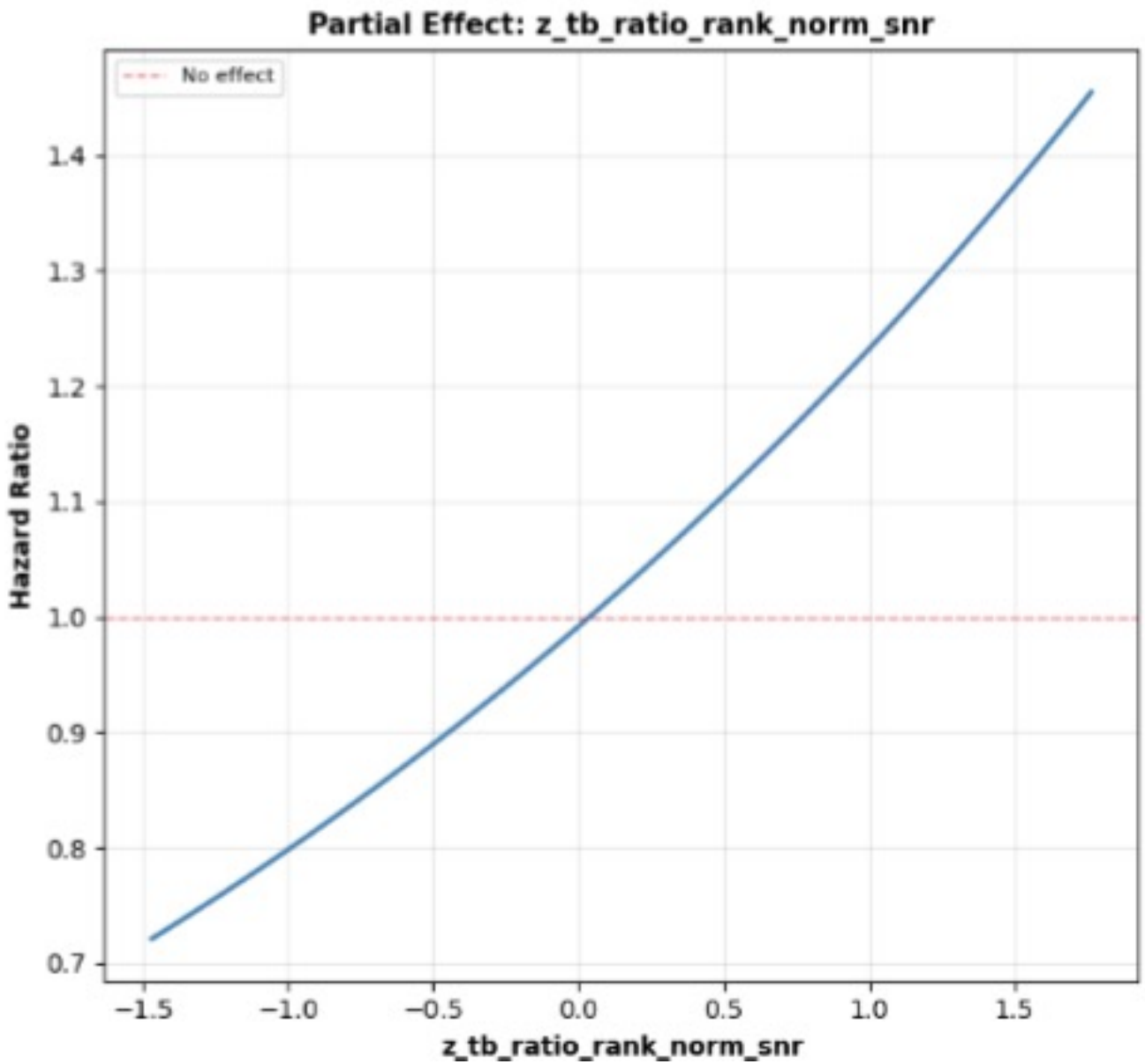
is about $-5.38\times$

the effect of A $+1\ \sigma$ change in $z_prestige_mean$

Since

$$\frac{b_{cum}}{b_{rank}} \approx -5.38$$

=== 📦 12.6. PARTIAL EFFECTS PLOTS ===



RUN 1 - Using

z_cum_tb_rcvd_ratio_snr
z_pool_tb_ratio_mean_snr
z_star_pool_interaction

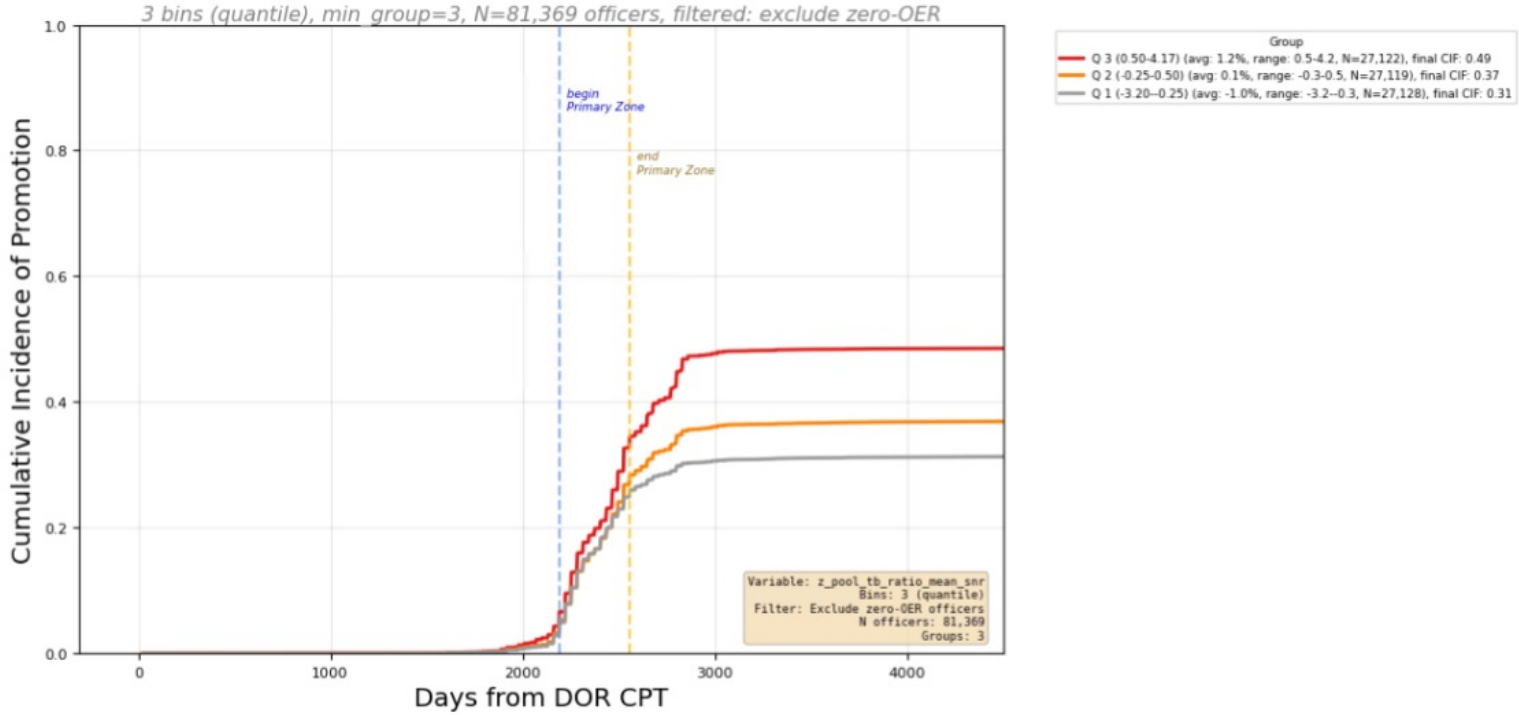
📈 CELL 11+: COX ANALYSIS & PLOTTING

Competing Risks

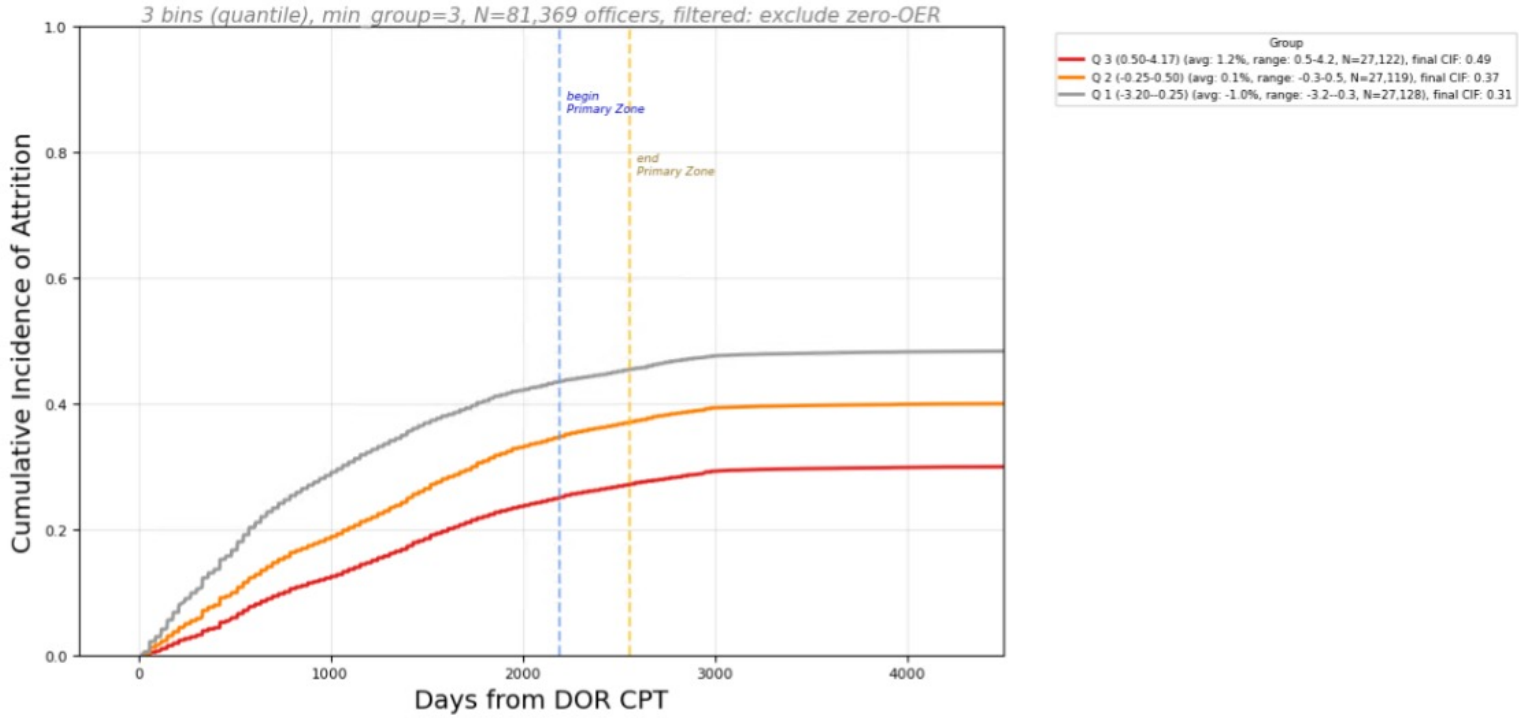
Cumulative Incidence Function Curve (empirical)

$z_pool_tb_ratio_mean_snr$

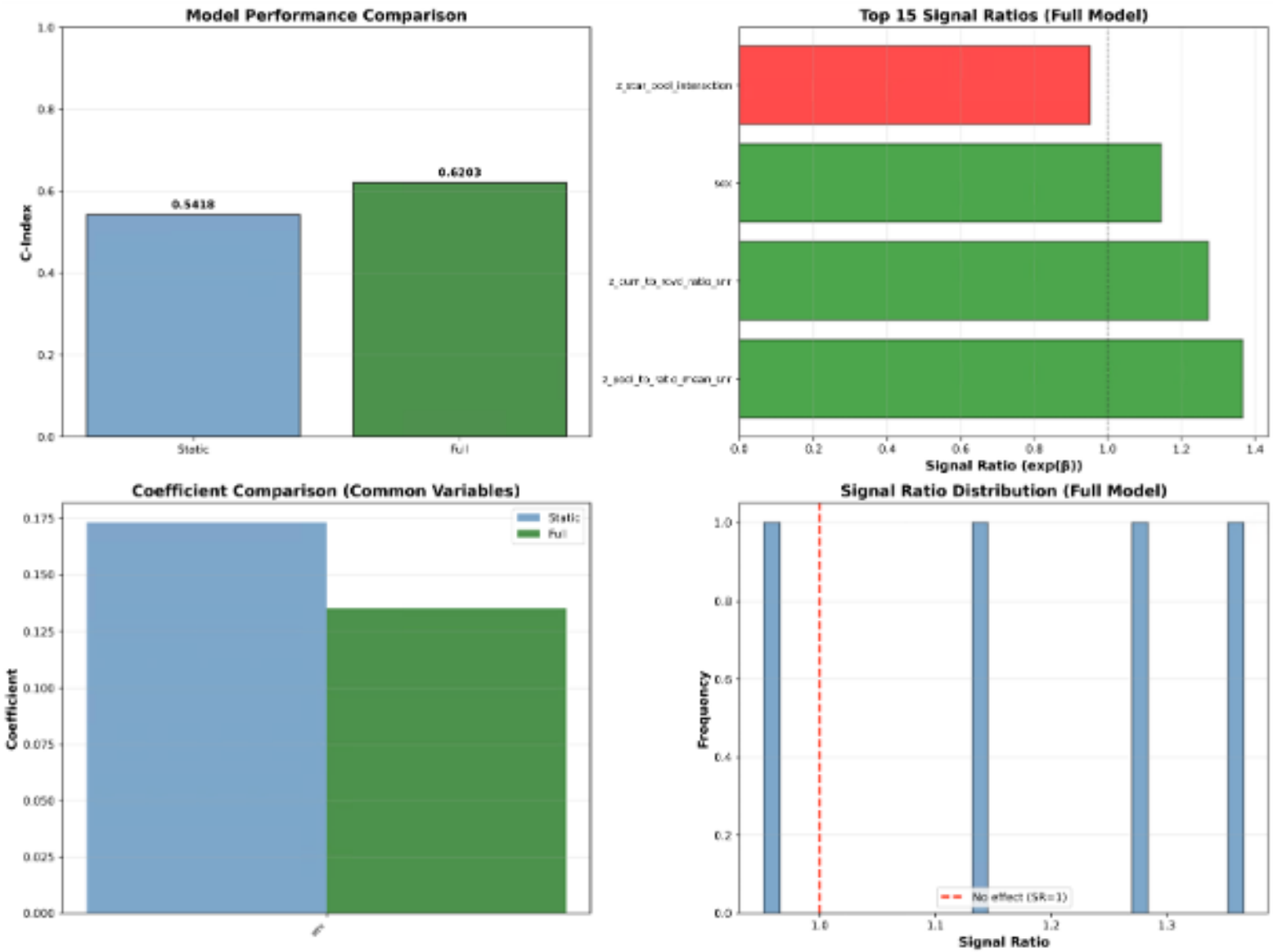
Promotion Probability (CIF): Competing Risks: Z Pool TB Ratio Mean SNR



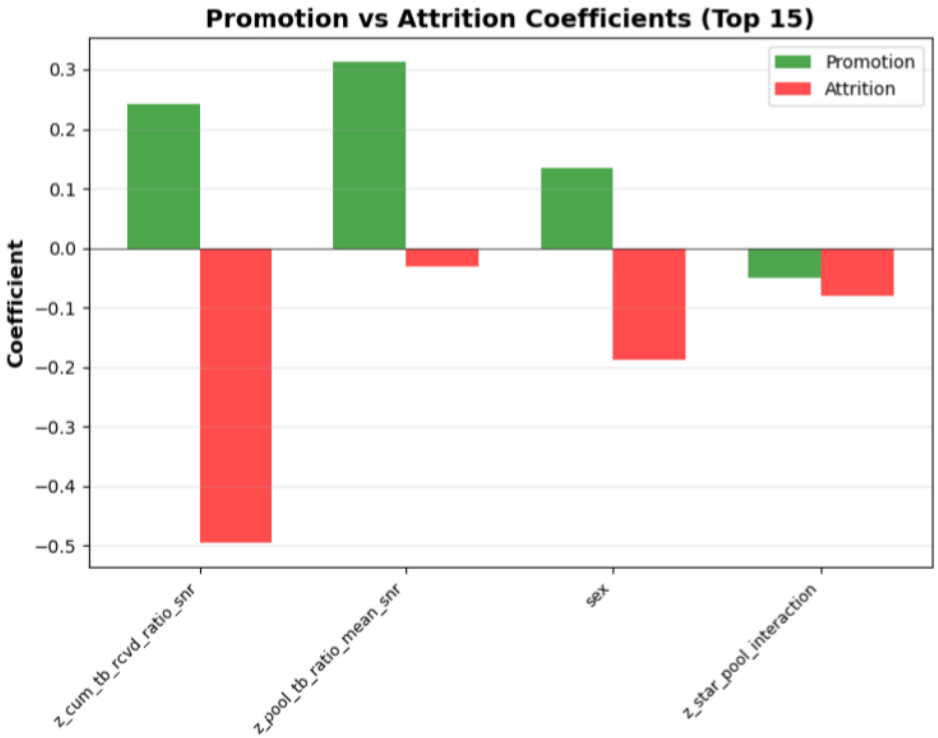
Attrition Probability (CIF): Competing Risks: Z Pool TB Ratio Mean SNR

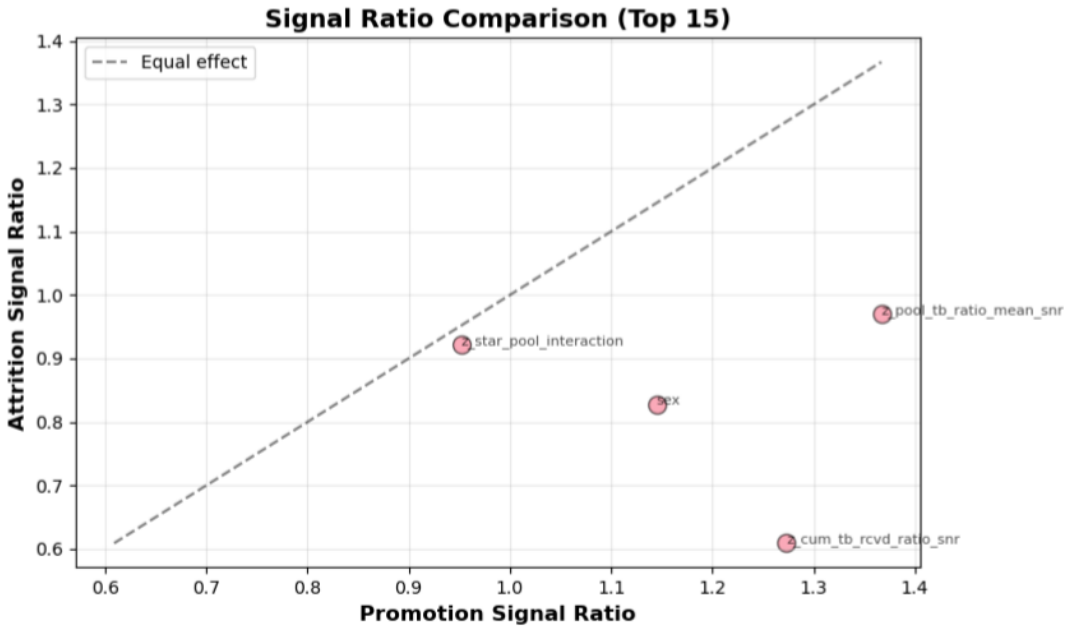


=== 📦 12.4. MODEL COMPARISON AND SIGNAL RATIOS ===

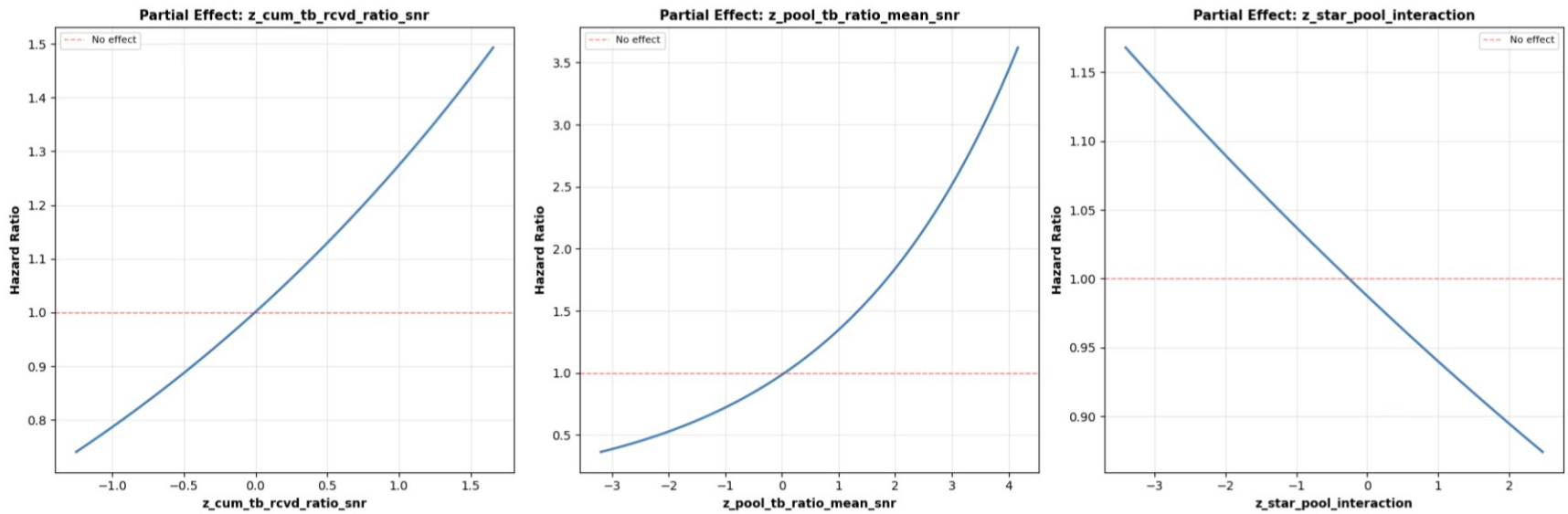


=== 📦 12.5. COMPETING RISKS ANALYSIS ===

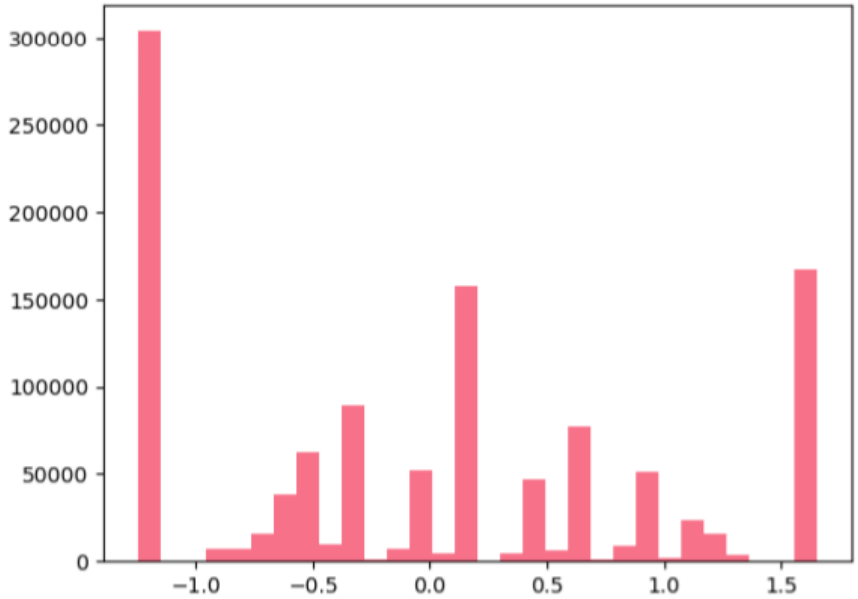




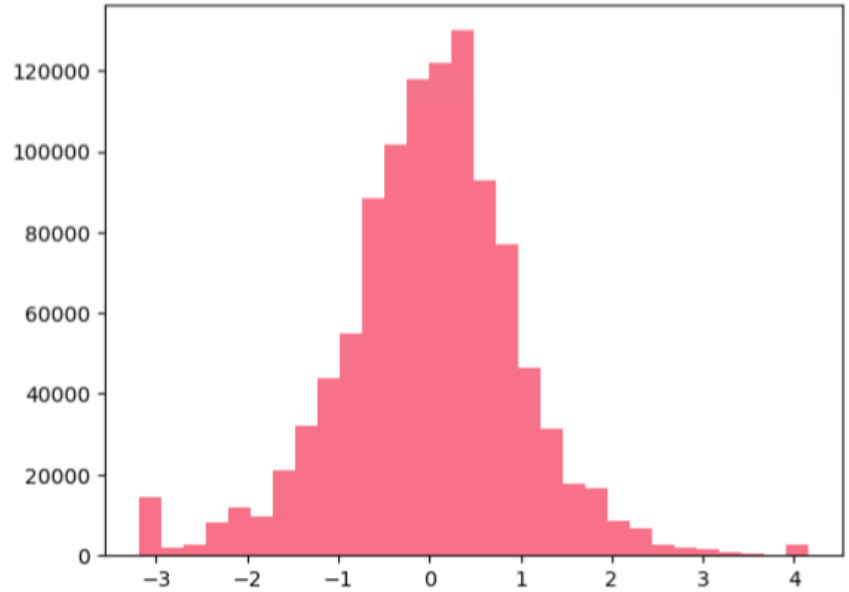
=== 📦 12.6. PARTIAL EFFECTS PLOTS ===



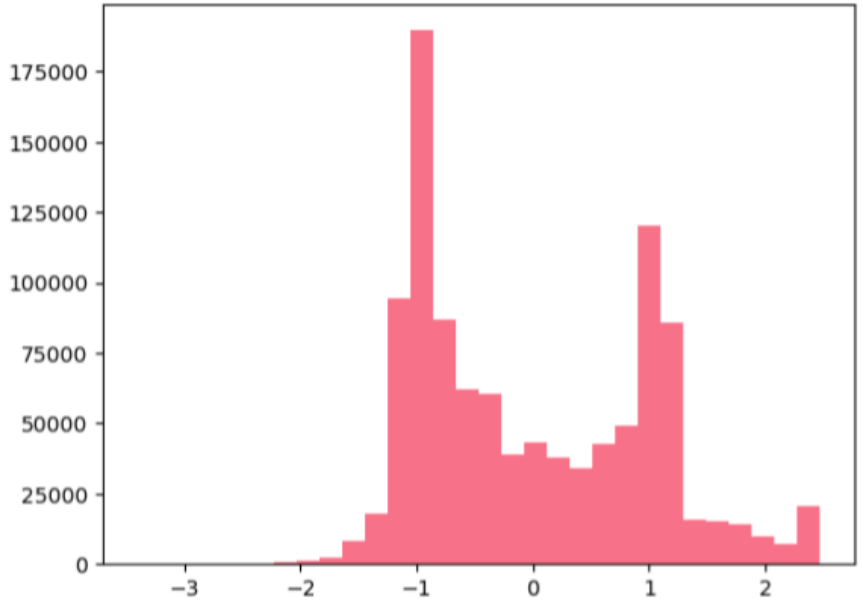
Column: z_cum_tb_rcvd_ratio_snr --> Mean: -0.00000000, Min: -1.25084007, Max: 1.65707204



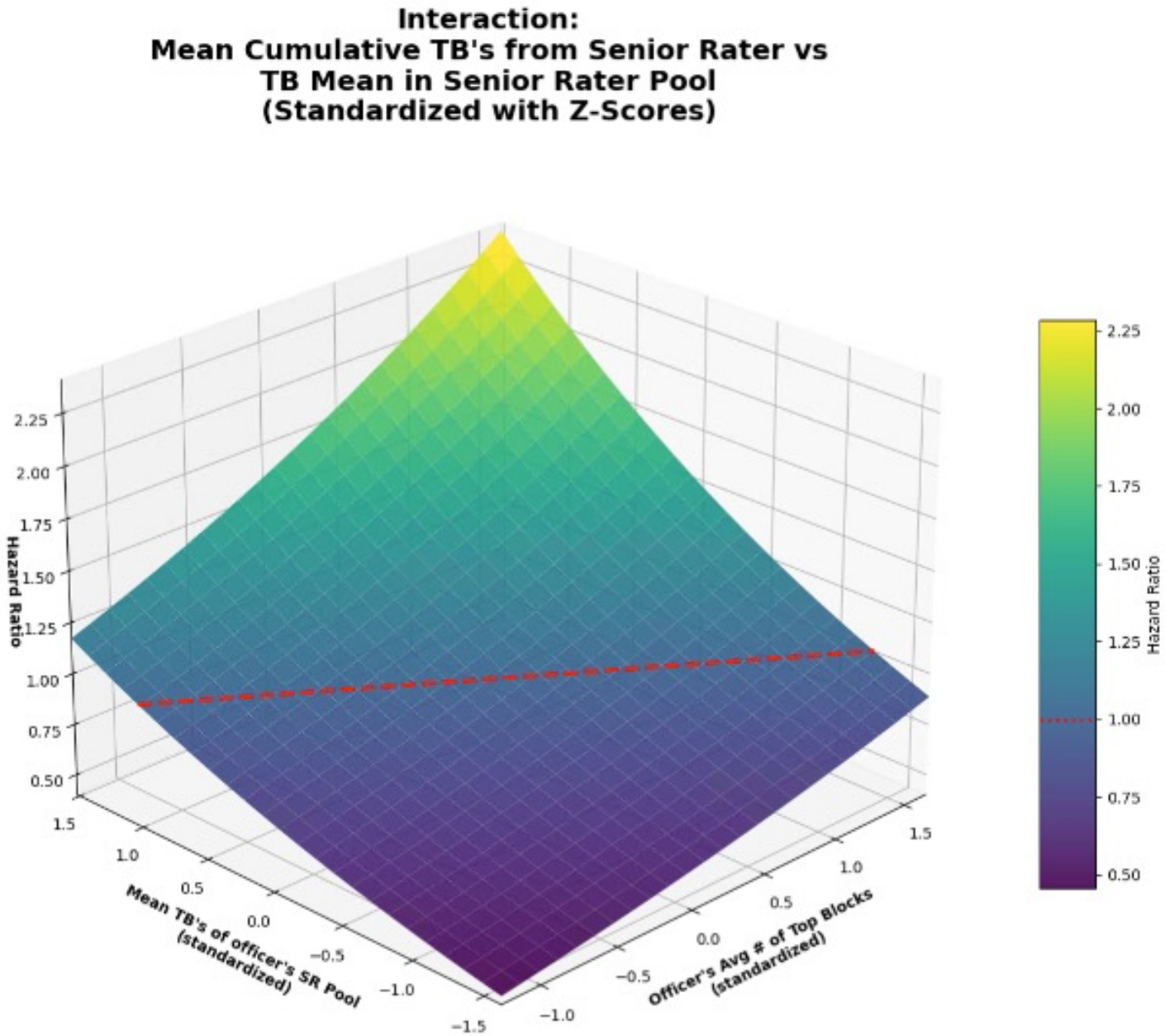
Column: z_pool_tb_ratio_mean_snr --> Mean: -0.00000000, Min: -3.19540035, Max: 4.16577111

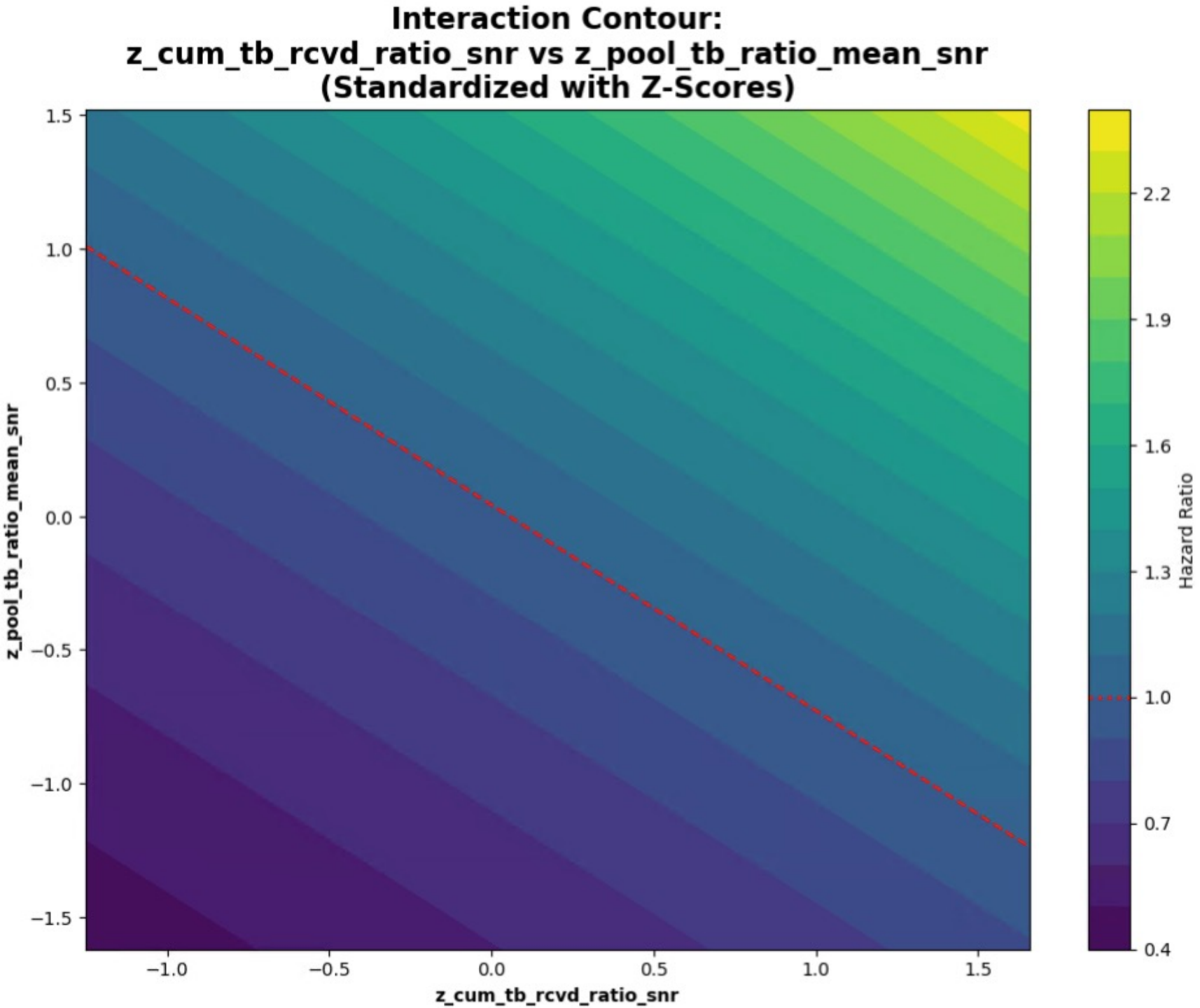


Column: z_star_pool_interaction --> Mean: -0.00000000, Min: -3.40592396, Max: 2.47318154



=== 📦 12.7. INTERACTION EFFECTS (3D PLOTS) ===




Interaction Contour:
z_cum_tb_rcvd_ratio_snr vs z_pool_tb_ratio_mean_snr
(Standardized with Z-Scores)
z_pool_tb_ratio_mean_snr
z_cum_tb_rcvd_ratio_snr
Hazard Ratio
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
2.2
1.9
1.6
1.3
1.0
0.7
0.4

=== 📦 12.3. FIT FULL MODEL (STATIC + TIME-VARYING) ===

A	B	C
variable	coefficient	signal_ratio
sex	0.1353158418089031	1.144898334012884
z_cum_tb_rcvd_ratio_snr	0.24128259915494218	1.2728806994954573
z_pool_tb_ratio_mean_snr	0.3124743979571531	1.3668029477782146
z_star_pool_interaction	-0.04931090064538285	0.95188514198453

To quantify the two effects with respect to each other:

If the **fitted model** is

$$\log(HR) = b1 * x + b2 * y + b3 * x * y$$

with $x = z_cum_tb_rcvd_ratio_snr$ and $y = z_pool_tb_ratio_mean_snr$,

then:

Marginal effect of x at a given y: $\frac{\partial}{\partial x}\log(HR) = b1 + b3 * y$

Marginal effect of y at a given x: $\frac{\partial}{\partial y}\log(HR) = b2 + b3 * x$

A simple **trade-off rate** (how much x you need to change to offset y) is therefore:

$$\frac{dx}{dy} = -\frac{(b2 + b3 * x)}{(b1 + b3 * y)}$$

If there's **no interaction** $(b3 = 0)$, then it simplifies to: $\frac{dx}{dy} = -\frac{b2}{b1}$

So, a $+1\ \sigma$ increase in $z_pool_tb_ratio_mean_snr$,

is equivalent to

a $-\frac{b2}{b1}\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$.

Using the generated coefficient table:

A	B	C
variable	coefficient	signal_ratio
sex	0.1353158418089031	1.144898334012884
z_cum_tb_rcvd_ratio_snr	0.24128259915494218	1.2728806994954573
z_pool_tb_ratio_mean_snr	0.3124743979571531	1.3668029477782146
z_star_pool_interaction	-0.04931090064538285	0.95188514198453

$b_{cum} = 0.3169$ for $z_cum_tb_rcvd_ratio_snr$

$b_{pool} = -0.0590$ for $z_pool_tb_ratio_mean_snr$,

With no interaction term, the trade-off at constant HR is:

$$\frac{dx}{dy} = -\frac{b_{pool}}{b_{cum}} = -1.2951$$

Interpretation (z-score units):

A $+1\ \sigma$ change in $z_pool_tb_ratio_mean_snr$,

is offset by

A $-1.2951\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$

to keep HR constant.

Equivalently:

A $+1\ \sigma$ change in $z_cum_tb_rcvd_ratio_snr$

is about $-0.7722\times$

the effect of A $+1\ \sigma$ change in $z_pool_tb_ratio_mean_snr$,

Since

$$\frac{b_{cum}}{b_{pool}} \approx -0.7722$$

RUN 2 - Using

z_tb_ratio_rank_norm_snr
z_pool_tb_ratio_mean_snr
z_rank_pool_interaction

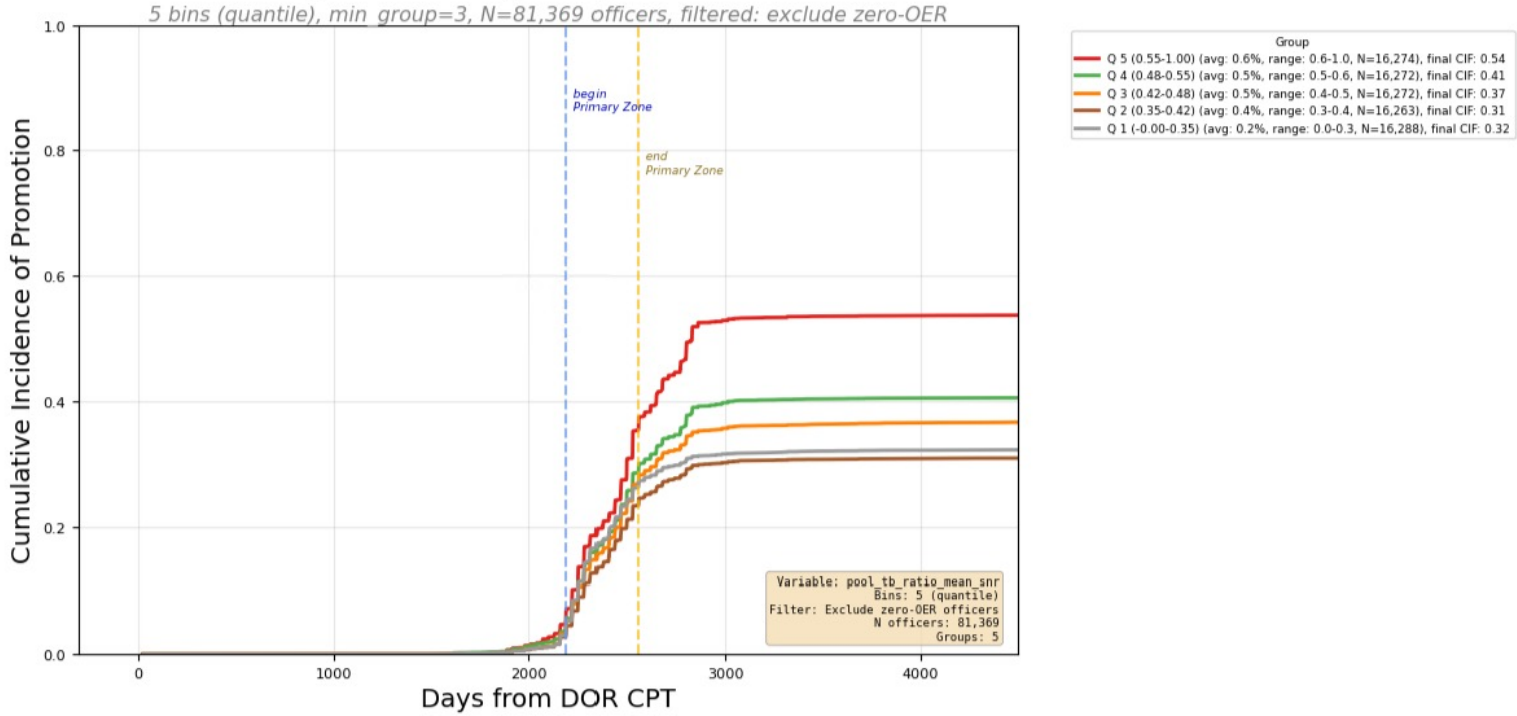
📈 CELL 11+: COX ANALYSIS & PLOTTING

Competing Risks

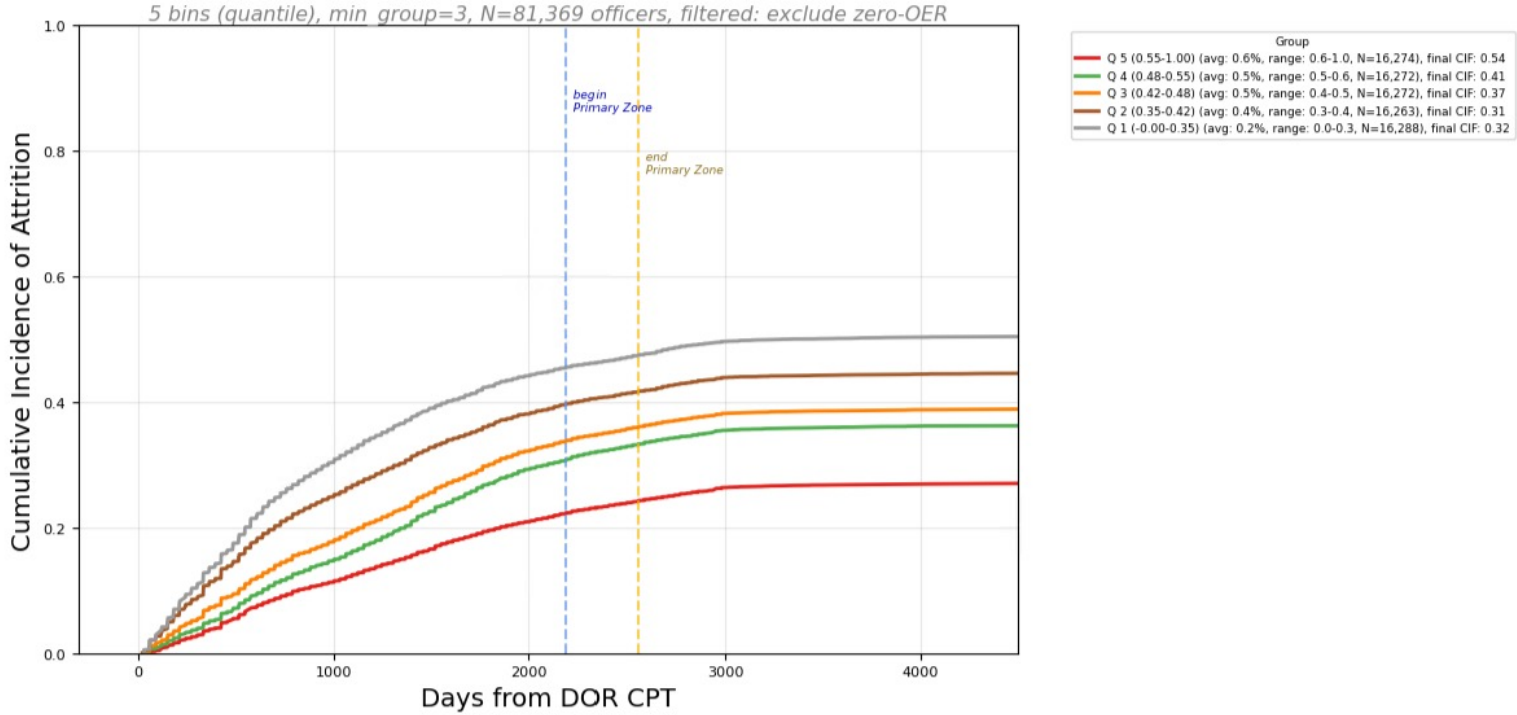
Cumulative Incidence Function Curve (empirical)

$z_pool_tb_ratio_mean_snr$

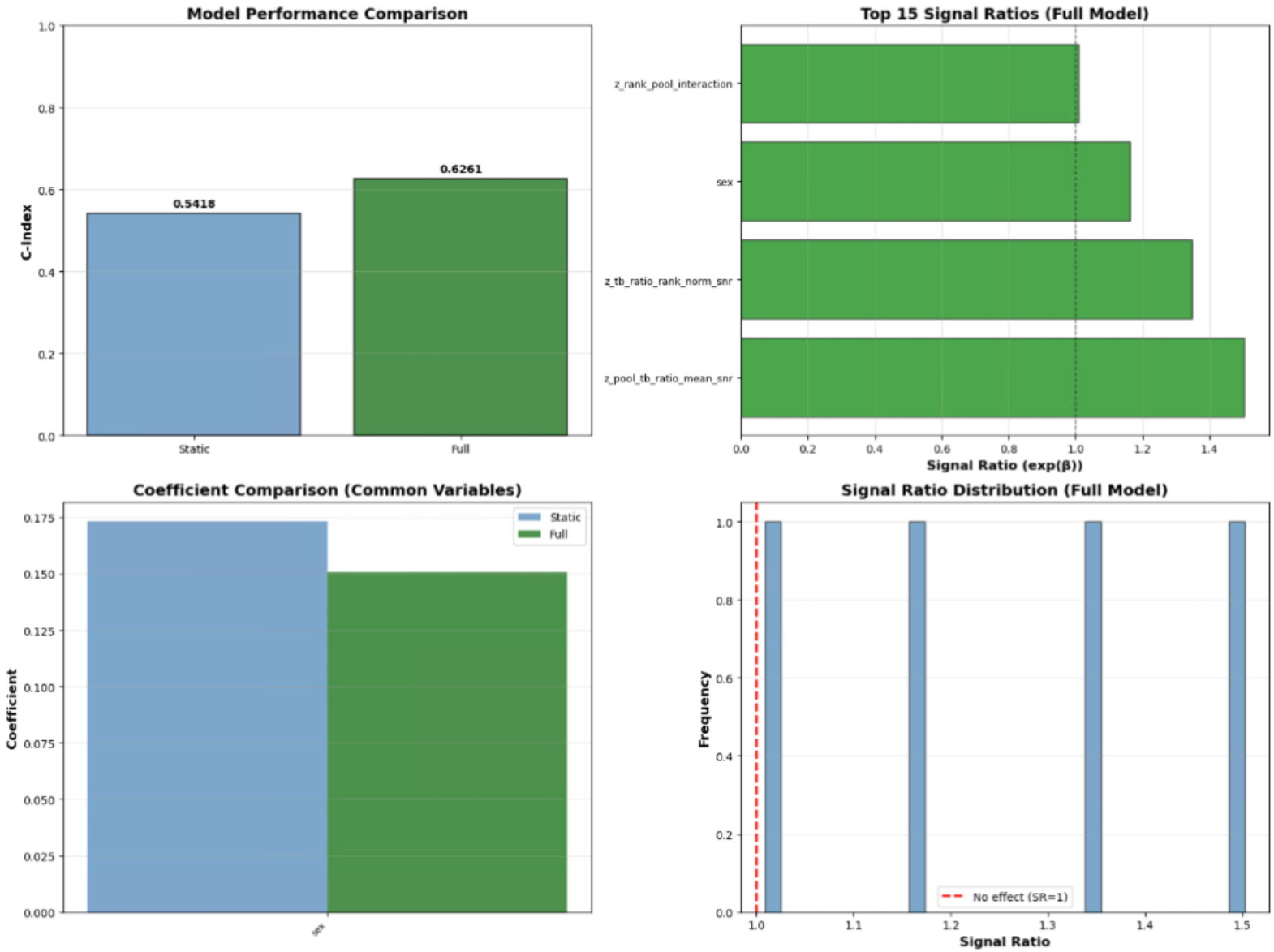
Promotion Probability (CIF): Competing Risks: Pool TB Ratio Mean SNR



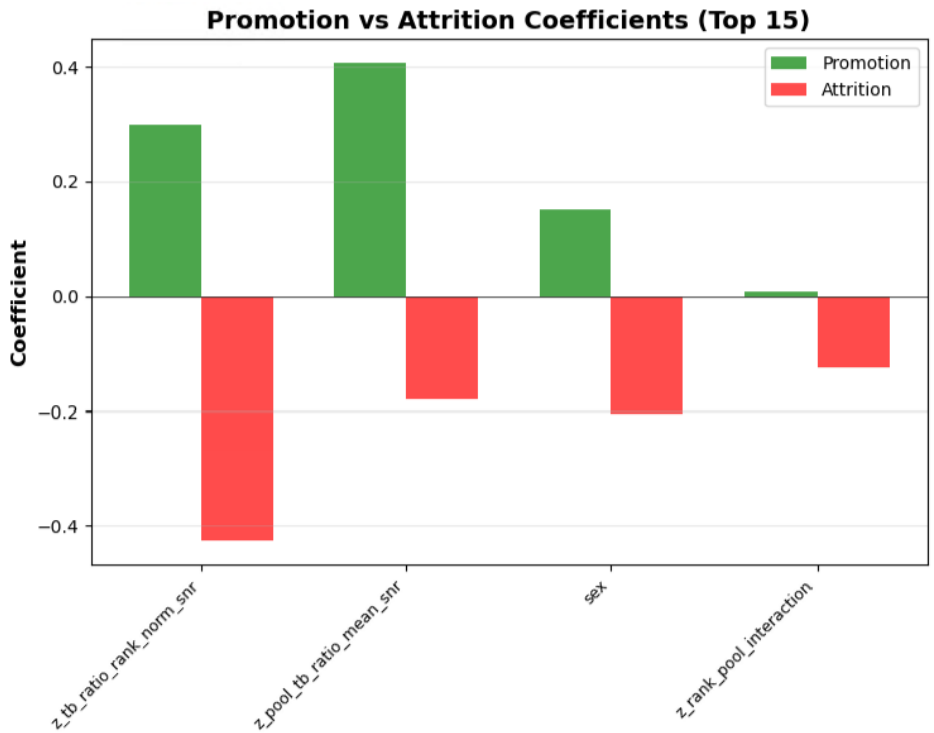
Attrition Probability (CIF): Competing Risks: Pool TB Ratio Mean SNR

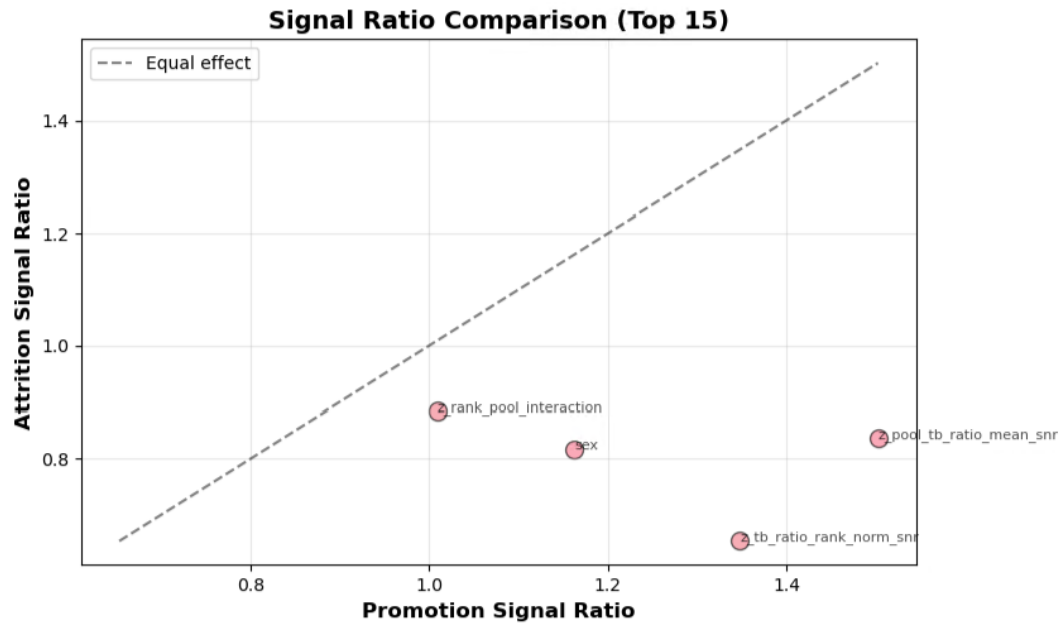


=== 📦 12.4. MODEL COMPARISON AND SIGNAL RATIOS ===

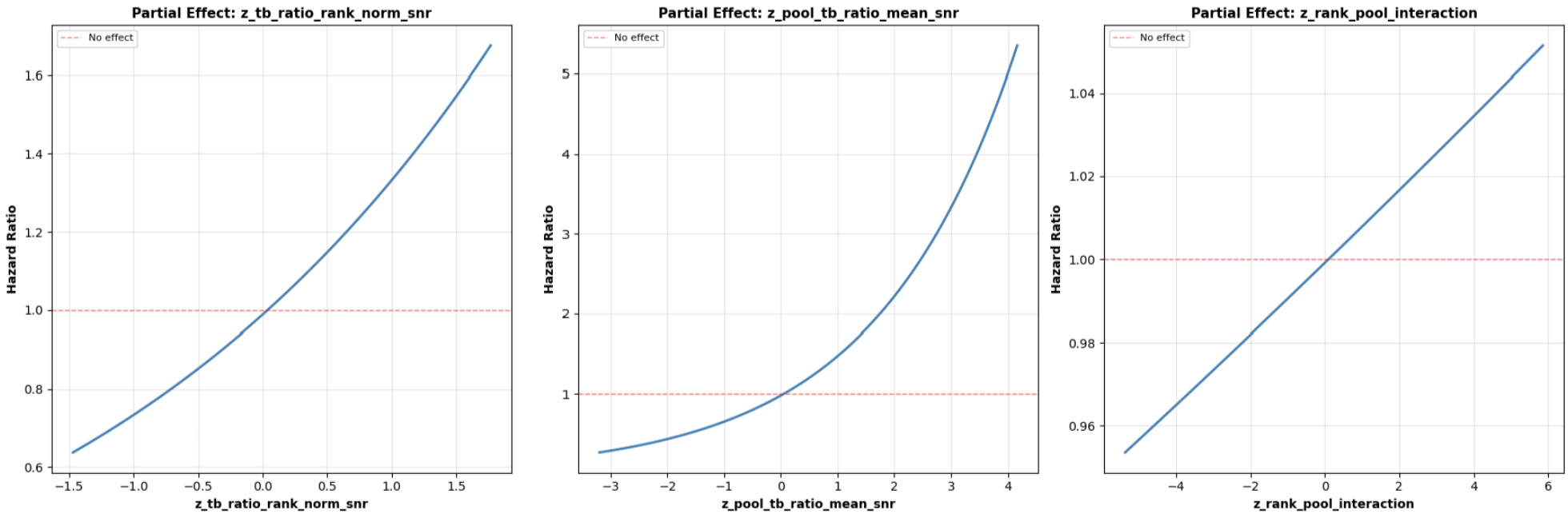


=== 📦 12.5. COMPETING RISKS ANALYSIS ===

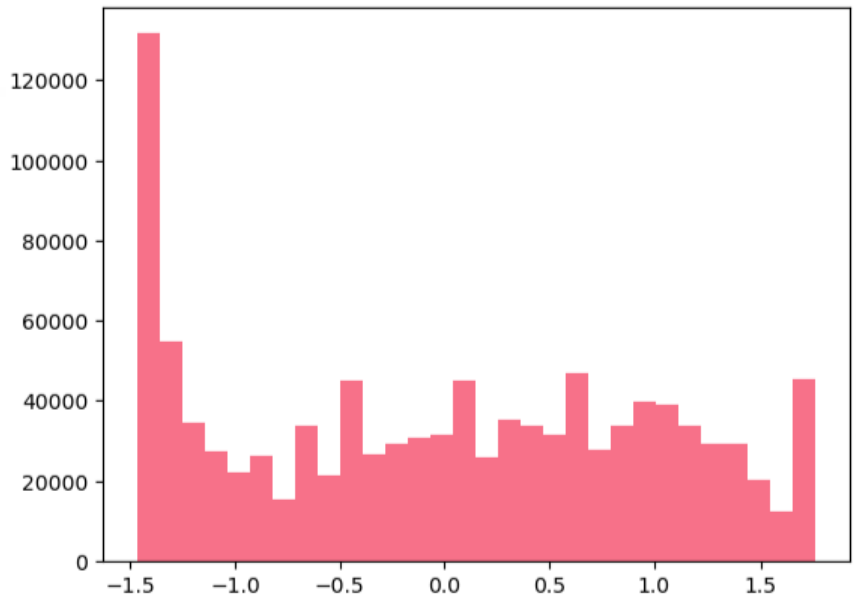




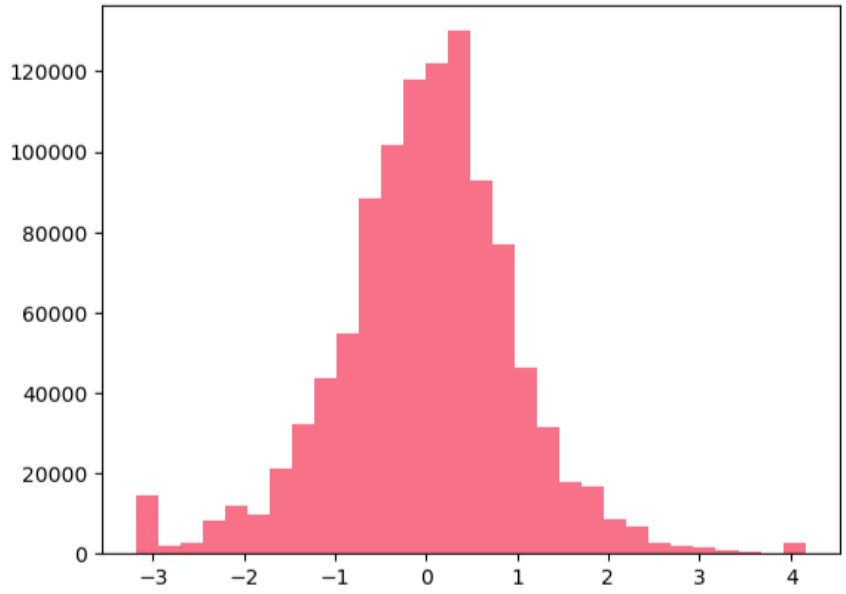
=== 📦 12.6. PARTIAL EFFECTS PLOTS ===



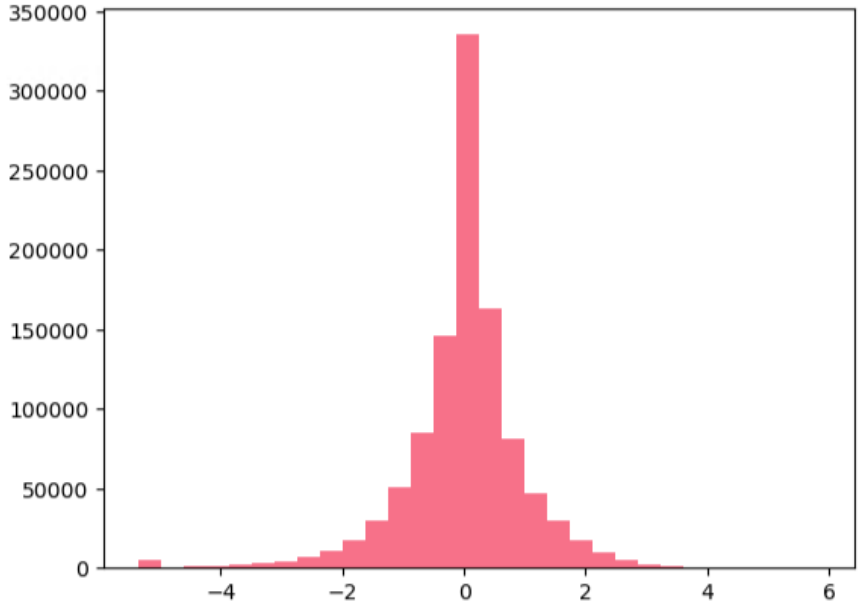
Column: z_tb_ratio_rank_norm_snr --> Mean: 0.00000000, Min: -1.46880511, Max: 1.76449261



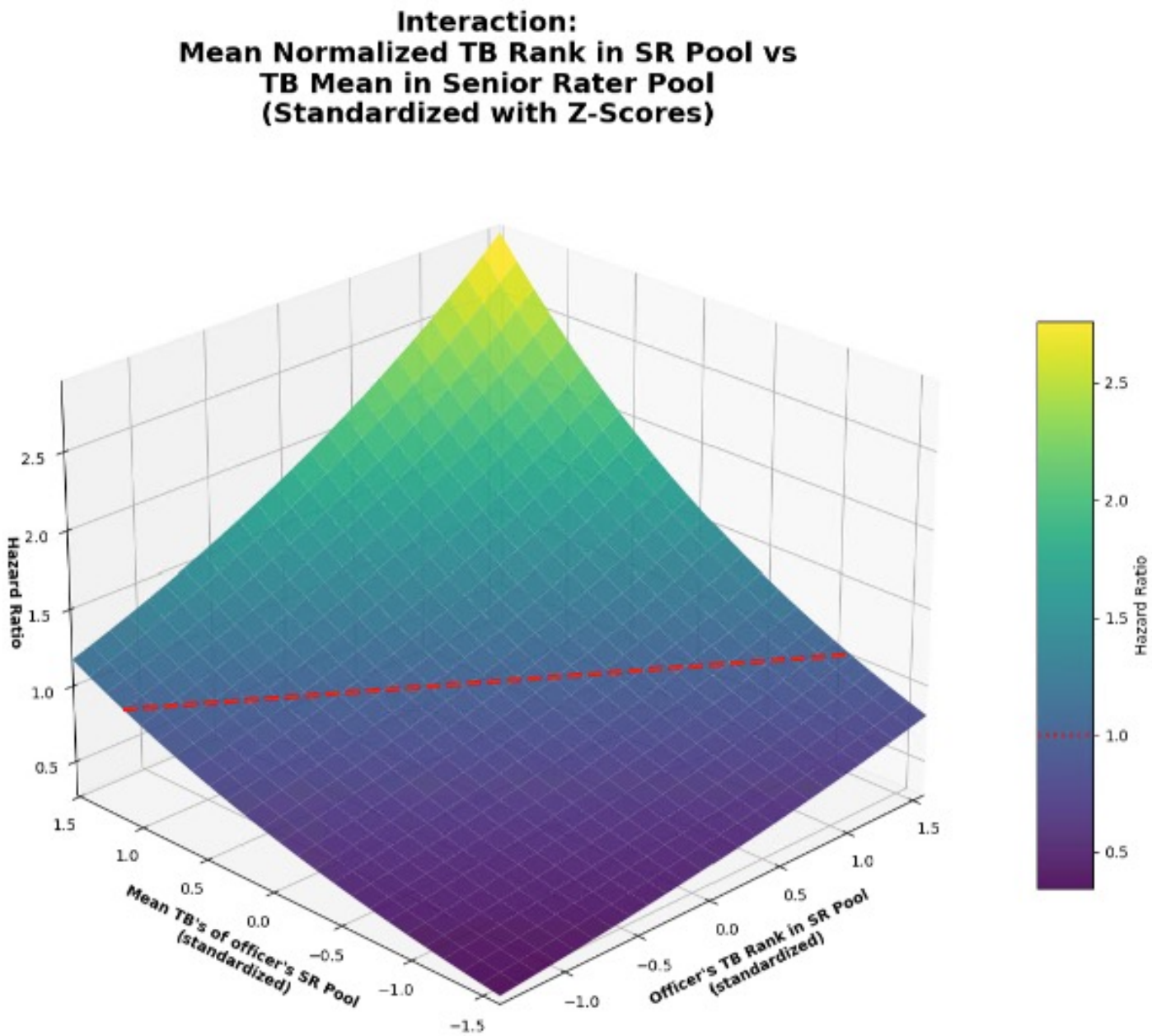
Column: z_pool_tb_ratio_mean_snr --> Mean: -0.00000000, Min: -3.19540035, Max: 4.16577111

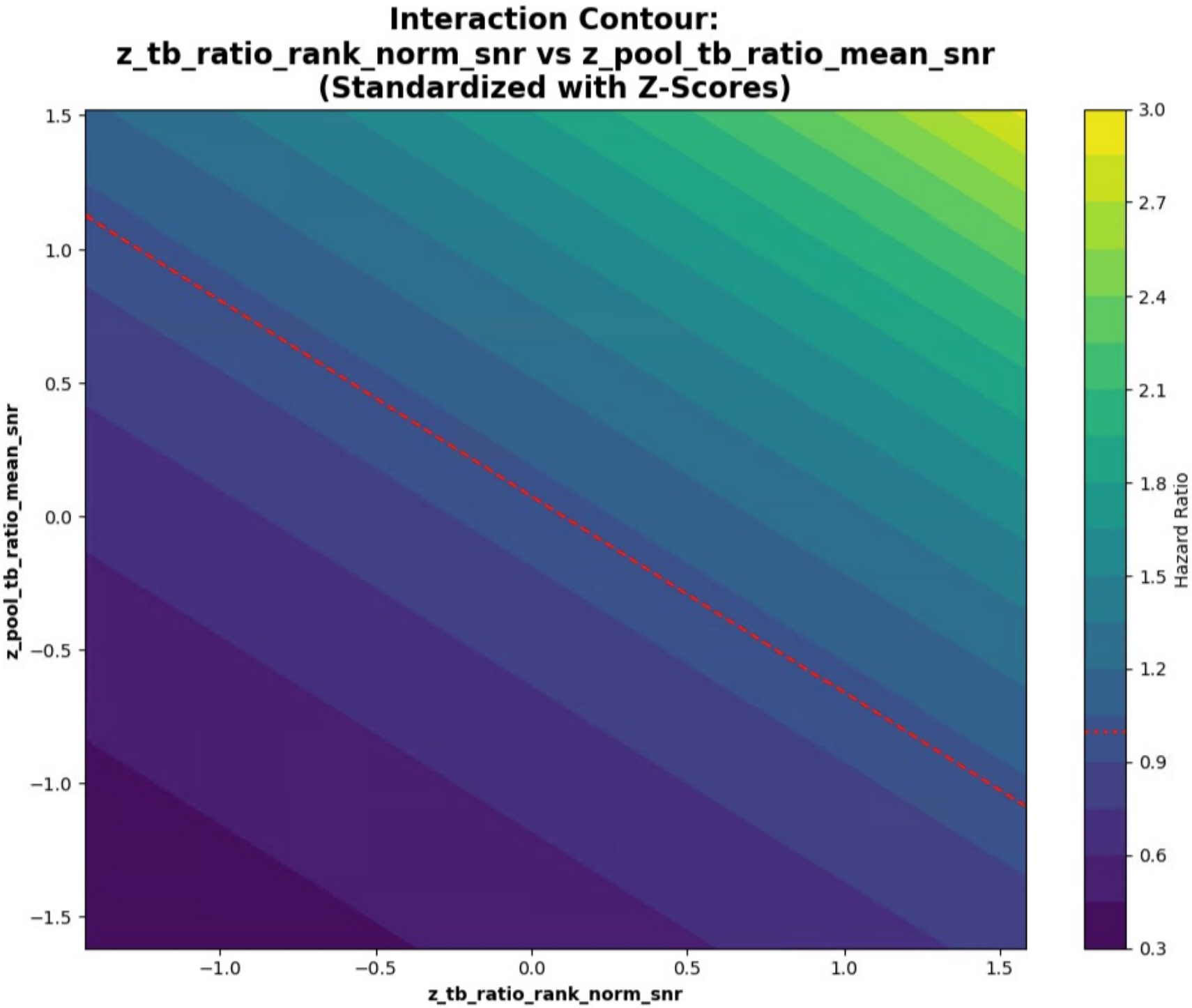


Column: z_rank_pool_interaction --> Mean: 0.00000000, Min: -5.36817229, Max: 5.85903192



=== 📦 12.7. INTERACTION EFFECTS (3D PLOTS) ===




Interaction Contour:
z_tb_ratio_rank_norm_snr vs z_pool_tb_ratio_mean_snr
(Standardized with Z-Scores)
z_pool_tb_ratio_mean_snr
z_tb_ratio_rank_norm_snr
Hazard Ratio
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
3.0
2.7
2.4
2.1
1.8
1.5
1.2
0.9
0.6
0.3

=== 📦 12.3. FIT FULL MODEL (STATIC + TIME-VARYING) ===

A	B	C
variable	coefficient	signal_ratio
sex	0.15052172986982648	1.1624405645108404
z_tb_ratio_rank_norm_snr	0.2987774118380368	1.3482094945982386
z_pool_tb_ratio_mean_snr	0.4072833773789217	1.5027298849870216
z_rank_pool_interaction	0.008697572773475376	1.008735506557074

To quantify the two effects with respect to each other:

If the **fitted model** is

$$\log(HR) = b1 * x + b2 * y + b3 * x * y$$

with $x = z_tb_ratio_rank_norm_snr$ and $y = z_pool_tb_ratio_mean_snr$,

then:

Marginal effect of x at a given y: $\frac{\partial}{\partial x}\log(HR) = b1 + b3 * y$

Marginal effect of y at a given x: $\frac{\partial}{\partial y}\log(HR) = b2 + b3 * x$

A simple **trade-off rate** (how much x you need to change to offset y) is therefore:

$$\frac{dx}{dy} = -\frac{(b2 + b3 * x)}{(b1 + b3 * y)}$$

If there's **no interaction** $(b3 = 0)$, then it simplifies to: $\frac{dx}{dy} = -\frac{b2}{b1}$

So, a $+1\ \sigma$ increase in $z_pool_tb_ratio_mean_snr$,

is equivalent to

a $-\frac{b2}{b1}\ \sigma$ change in $z_tb_ratio_rank_norm_snr$.

Using the generated coefficient table:

A	B	C
variable	coefficient	signal_ratio
sex	0.15052172986982648	1.1624405645108404
z_tb_ratio_rank_norm_snr	0.2987774118380368	1.3482094945982386
z_pool_tb_ratio_mean_snr	0.4072833773789217	1.5027298849870216
z_rank_pool_interaction	0.008697572773475376	1.008735506557074

$b_{rank} = 0.3169$ for $z_tb_ratio_rank_norm_snr$.
$b_{pool} = -0.0590$ for $z_pool_tb_ratio_mean_snr$

With no interaction term, the trade-off at constant HR is:

$$\frac{dx}{dy} = -\frac{b_{pool}}{b_{rank}} = -1.3632$$

Interpretation (z-score units):

A $+1\ \sigma$ change in $z_pool_tb_ratio_mean_snr$
is offset by
A $+0.1863\ \sigma$ change in $z_tb_ratio_rank_norm_snr$
to keep HR constant.

Equivalently:

A $+1\ \sigma$ change in $z_tb_ratio_rank_norm_snr$

is about $-0.7336\times$

the effect of A $+1\ \sigma$ change in $z_pool_tb_ratio_mean_snr$

Since

$$\frac{b_{rank}}{b_{pool}} \approx -0.7336$$